BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING

The Presence Bottleneck: Domain Adaptation and Low-End Deployment of SAM 3

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Declaration of Candidate

This is to certify that the work presented in this thesis is the outcome of the analysis and experiments carried out by **Fairuz Khan**, **Rifat Sarwar** and **Wasif Uddin** under the supervision of **Dr. Md. Hasanul Kabir**, [Designation], Department of Computer Science and Engineering, Islamic University of Technology, Dhaka, Bangladesh. It is also declared that neither this thesis nor any part of it has been submitted anywhere else for any degree or diploma. Information derived from the published and unpublished work of others has been acknowledged in the text and a list of references is given.

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Dedicated to our families and mentors, whose unwavering support guided this research journey

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Abstract

SAM 3 segments every instance of a concept named in text, but its accuracy drops on imagery unlike its training data, and with 840 M parameters it is too large for low-end hardware. In the preliminary phase of this thesis we proposed two remedies: *ConceptBank*, which steers SAM 3's text embeddings toward visual prototypes mined from target-domain images and appeared to improve LoveDA Urban by +0.019 mIoU, and implicit-depth and deformable-attention architectures for efficiency. This thesis re-evaluates both under a rigorous protocol—frozen splits, a pre-registered primary endpoint, image-level bootstrap intervals and matched random controls—and follows the evidence to what actually works.

For domain adaptation, the preliminary gain does not reproduce (+0.0001, 95% CI $[-0.003, +0.003]$). The preliminary evaluation compared one instance mask with a whole class region, which halved SAM 3's measured accuracy (0.245 versus 0.503); the prototypes are 17× smaller than the text embeddings and retrieved at chance; and correct, shuffled and absent prototypes give identical results. What remains is a data-free rebalancing of each prompt's token-mean "gist" against its per-token "detail". Fixed on aerial data and applied unchanged, it improves underwater segmentation by +0.090 $[+0.024, +0.161]$ and beats 19 random directions; on aerial imagery its reliable gain is about +0.02, and on natural photographs there is none. The gain consists of objects SAM 3 had localised but vetoed through a pessimistic prompt-level *presence* score.

For efficiency, a measured cost census shows that the preliminary architectural proposals target 0.1–4% of compute. INT8 quantization with a cached vocabulary runs SAM 3 in a 657 MB file and 2.5 GB of RAM with no measurable accuracy loss on 400 unseen COCO images (0.810 versus 0.809 mIoU). We measure EfficientSAM3, the only released low-end SAM 3, for the first time and trace its accuracy loss to presence collapsing under feature distillation. Sharing per-concept computation fails for the same reason. Distilling the small image backbone *through* SAM 3's frozen concept path on unlabelled images, with teacher-balanced concept sampling, raises it on a 400-image confirmation set from 0.339 to 0.486 mIoU (+0.147 $[+0.113, +0.184]$) at unchanged size and speed, and a matched feature-imitation control attributes the gain to the output-level objective (+0.265).

Across both problems one component decides success or failure: SAM 3's presence score.

Keywords: SAM 3, promptable concept segmentation, domain shift, knowledge distillation, quantization, reproducibility.

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List of Abbreviations

BCE	Binary Cross-Entropy
CI	Confidence Interval
CPU / GPU	Central / Graphics Processing Unit
DEQ	Deep Equilibrium Model
DETR	DEtection TRansformer
FLOPs	Floating-Point Operations
FP / FN / TP	False Positive / False Negative / True Positive
FPN	Feature Pyramid Network
INT8	8-bit Integer (quantization)
IoU / mIoU	Intersection over Union / mean IoU
MDE	Minimum Detectable Effect
MDEQ	Multiscale Deep Equilibrium Model
OOD	Out-of-Distribution
PCS	Promptable Concept Segmentation
RevDEQ	Reversible Deep Equilibrium Model
RSS	Resident Set Size (process memory)
SAM	Segment Anything Model
SUIM	Segmentation of Underwater IMagery (dataset)
ViT	Vision Transformer

Chapter 1

Introduction

Intelligent vision systems depend on understanding images at the level of individual pixels. Image segmentation partitions an image into meaningful regions and labels them, which is what autonomous navigation, medical image analysis, environmental monitoring from aerial imagery and robotic manipulation require. Over the last decade segmentation has moved from task-specific convolutional networks trained for a fixed label set [26, 39] to *foundation models* that respond to prompts. The Segment Anything Model (SAM) [23] segments an object given a point or a box; SAM 2 [36] extends this to video; and SAM 3 [8] introduces *promptable concept segmentation*: given a short noun phrase such as "building" or "diver", it returns a mask for every instance of that concept in the image.

This thesis is about the gap between what SAM 3 can do on a benchmark of ordinary photographs and what is needed to use it in practice.

1.1 Motivation

Two obstacles stand between SAM 3 and many practical uses.

Domain shift. SAM 3 was trained mostly on ordinary photographs. On imagery with a different geometry or appearance—nadir aerial tiles, where a building is a rooftop seen from directly above, or turbid, colour-cast underwater scenes—its accuracy drops. Supervised fine-tuning recovers accuracy but requires labelled target-domain data, which is expensive in exactly the domains where it is needed.

Computational cost. SAM 3 has 840 M parameters. Measured on a desktop CPU, one image with five text prompts takes about 17 s, and building the model needs close to 8 GB of memory. Low-end hardware—a laptop CPU, an embedded board, a drone—cannot run it as released.

Both obstacles invite *inference-time* remedies that avoid labels and retraining. Such remedies are attractive, but small reported gains from them are easy to produce and hard to verify: a threshold, a read-out rule or the choice of evaluation images can create or hide an

effect of the same size as the remedy itself. This thesis is as much about measuring these remedies correctly as about the remedies.

1.2 From a Preliminary Proposal to This Thesis

In the preliminary phase of this project [22] we proposed one remedy for each obstacle (Chapter 3):

1. **ConceptBank**, a parameter-free domain adaptation method that clusters SAM 3's image features from target-domain support images into visual prototypes and steers the text embedding of each prompt toward them. On ten LoveDA Urban images it appeared to raise mIoU from 0.2373 to 0.2563 (+0.019), and to recover the forest class from 0.000.
2. **Architectural efficiency**, by replacing stacked transformer layers with deep equilibrium models (DEQ, MDEQ, RevDEQ) [1, 2] and global cross-attention with deformable attention [53].

Neither was tested under conditions that could have falsified it. The ConceptBank result was tuned and reported on the same ten images, under a single read-out rule and without controls; the original implementation was not preserved. The efficiency proposals were chosen by counting transformer layers rather than by measuring where the computation goes, and were never trained. This thesis therefore starts by rebuilding both from scratch inside one evaluation harness, re-testing them under a pre-registered protocol, and then pursuing what the evidence supports.

1.3 Research Questions

RQ1 Does ConceptBank's reported improvement hold under a correct evaluation protocol, with matched controls, on held-out images?

RQ2 If any effect of text-embedding steering exists, what is it, how large is it once every selection step is accounted for, and why does it help on some domains and not others?

RQ3 Where does SAM 3's computational cost actually lie, and do the preliminary architectural proposals target it?

RQ4 How accurate is the existing low-end SAM 3 (EfficientSAM3), which component loses its accuracy, and can SAM 3 itself be made to fit low-end memory budgets?

RQ5 Can a small image backbone be trained, without labels, to recover a substantial part of SAM 3's accuracy at unchanged cost?

1.4 Contributions

1. **A controlled re-evaluation of ConceptBank** (Chapter 5). The preliminary gain does not reproduce (+0.0001, a well-powered null). The preliminary protocol understated SAM 3 by half (0.245 versus 0.503). The method reads ground-truth labels, its prototypes are 17× smaller than text embeddings, and correct, shuffled and absent prototypes perform identically.
2. **A data-free gist/detail recalibration, sized honestly** (Chapter 6). Out of sample on underwater imagery it improves mIoU by +0.090 [+0.024, +0.161] and beats 19 random directions; on aerial imagery its reliable gain is about +0.02; on natural photographs it is zero.
3. **A mechanism with a confirmed prediction** (Chapter 6). The gain is detections SAM 3 had localised but vetoed through low *presence*; on COCO, where presence is saturated, nothing is rescued, as predicted.
4. **A measured account of SAM 3's cost and a low-memory deployment** (Chapter 7). A parameter and FLOP census shows the preliminary proposals target 0.1–4% of compute. INT8 SAM 3 with a cached vocabulary runs in 2.5 GB with no measurable accuracy loss (0.810 versus 0.809 on 400 unseen images). EfficientSAM3 is measured for the first time, and its loss is traced to presence collapse in its distilled image encoder; sharing per-concept computation fails for the same reason.
5. **Concept-in-the-loop output distillation** (Chapter 8). Training a small backbone through SAM 3's frozen concept path on unlabelled images, with teacher-balanced concept sampling, lifts it from 0.339 to 0.486 mIoU on 400 unseen COCO images (+0.147 over the released EfficientSAM3) at identical size and speed; a matched control attributes the gain to the objective.
6. **A reusable evaluation framework and a record of what failed** (Chapters 4 and 9). Frozen splits, raw per-image records, matched-norm controls and selection-aware intervals, together with three of our own intermediate claims that did not survive them.

1.5 Thesis Organisation

Chapter 2 reviews segmentation foundation models, SAM 3's architecture, domain adaptation and efficient models. Chapter 3 summarises the preliminary phase. Chapter 4 describes the evaluation framework. Part I (domain adaptation) comprises Chapter 5, which re-evaluates ConceptBank, and Chapter 6, which characterises the recalibration that remains and its mechanism. Part II (low-end deployment) comprises Chapter 7, on cost, EfficientSAM3, INT8 deployment and sharing, and Chapter 8, on output distillation. Chapter 9 draws the two parts together, discusses ethical considerations and limitations, and Chap-

ter 10 concludes. The appendices contain proofs, engineering pitfalls and reproduction instructions.

Chapter 2

Background and Related Work

This chapter introduces the models and ideas the thesis builds on: the evolution toward promptable segmentation, the components of SAM 3 that turn out to matter, how foundation models are adapted to new domains, the geometry of learned embeddings, efficient segmentation models, and the methodology of reproducible evaluation.

2.1 From Task-Specific Networks to Promptable Segmentation

Fully convolutional networks [26] made dense prediction trainable end to end, and encoder–decoder designs with skip connections such as U-Net [39] recovered the spatial detail lost to downsampling. These networks segment a fixed label set they were trained on. Transformers changed the formulation: DETR [7] casts detection as set prediction with a fixed number of learned object queries matched to ground truth, and MDETR [21] conditions the queries on free text, so that the set of detectable things is defined by language rather than by a closed vocabulary.

The Segment Anything Model [23] introduced *promptable* segmentation: a heavy image encoder runs once per image, and a light decoder returns masks for geometric prompts (points, boxes, masks). SAM 2 [36] added a streaming memory for video. SAM 3 [8] combines these lines into *promptable concept segmentation* (PCS): given a noun phrase, it detects, segments and (in video) tracks *all* instances of the concept.

2.2 SAM 3: The Components That Matter Here

Figure 2.1 summarises SAM 3's image path. All statements in this section were verified against the released source code.

Text features. A noun phrase is tokenised to 32 positions and encoded by a 354 M parameter transformer whose output is linearly resized from width 1024 to 256. The result,

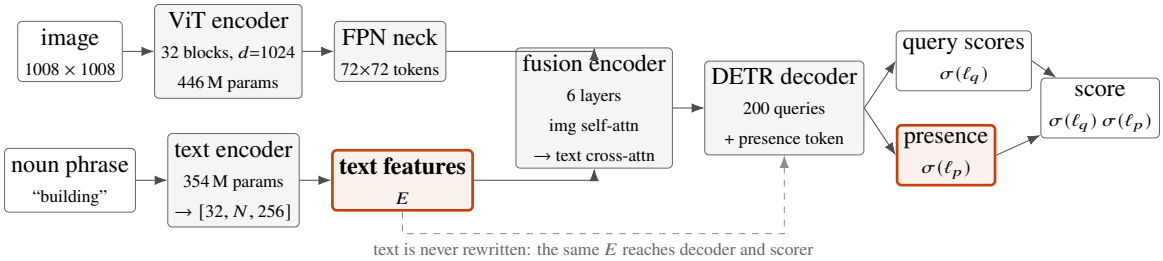


Figure 2.1: SAM 3's concept path. The ViT and neck run once per image; the text encoder, fusion encoder, decoder and mask head run once per concept. Every detection's score is the product of a per-query probability and a single per-prompt *presence* probability (highlighted), the component this thesis finds to be fragile. The text features E are the interception point for all embedding operators in Part I.

$E \in \mathbb{R}^{T\times N\times 256}$ with T=32 tokens and N prompts, comes with a padding mask (in the source, `True` marks padding).

One-way fusion. Each of the six fusion-encoder layers applies self-attention over the $72 \times 72 = 5184$ image tokens, cross-attention from image tokens to text tokens, and a feed-forward block. The encoder returns the text tokens unchanged, so an edit to E propagates unmodified to every downstream consumer: fusion cross-attention, the decoder, the scorer and the mask head.

Presence-factored scoring. The DETR decoder refines 200 learned object queries plus one *presence token*. Query i yields a classification logit from a dot-product scorer,

$$\ell_{q,i} = \frac{1}{\sqrt{d}} \langle W_h h_i,\ W_p\, \mathrm{pool}(E)\rangle, \qquad \mathrm{pool}(E) = \frac{1}{|\mathcal{V}|}\sum_{t\in\mathcal{V}} E_t, \tag{2.1}$$

where $\mathcal{V}$ indexes non-padding tokens, and the presence token yields one logit ℓ_p per prompt. The score of every detection is

$$s_i = \sigma(\ell_{q,i}) \cdot \sigma(\ell_p). \tag{2.2}$$

Presence is therefore a *multiplicative, prompt-level veto*: if $\sigma(\ell_p)$ is low, every instance of the concept is suppressed below any reasonable threshold, however well it was localised.

Where the parameters are. The image encoder holds 446 M of 840 M parameters and the text encoder 354 M (42%); the fusion encoder, decoder, heads and neck together hold under 5%. Section 7.1 measures where the *computation* is.

2.3 Adapting Foundation Models to New Domains

SAM-family models degrade on imagery far from their training distribution, including remote sensing [34] and underwater scenes [19]. The dominant remedy is supervised parameter-efficient fine-tuning: LoRA-style adapters and reference-guided variants reach about 52–53% mIoU on LoveDA [46] with a few hundred thousand trainable parameters [4, 9, 16]. Test-time adaptation [44] and prompt learning [15, 52] adapt parameters or embeddings with gradients. Training-free open-vocabulary methods usually build prototypes *in visual feature space* and apply them at the output [40, 45]. ConceptBank, our preliminary method, instead injected visual prototypes into the *text* embedding space.

2.4 Geometry of Learned Embeddings

Contextual and contrastive embeddings are strongly anisotropic, occupying a narrow cone [13], and image and text embeddings of jointly trained models occupy separate regions, the *modality gap* [14, 24]. Removing the common mean and dominant principal directions improves word and sentence embeddings [30, 38]. Both facts become important in Chapters 5 and 6.

2.5 Efficient Segmentation Models

Efficient SAM variants. Efficiency work on SAM is surveyed in [41]. For SAM 3, EfficientSAM3 [49] distils the image encoder into EfficientViT [6], RepViT [43] and TinyViT [48] students and the text encoder into MobileCLIP [42]; its report defers all accuracy and latency results to a future revision. SAM3-LiteText [50] distils only the text encoder. EdgeSAM [51] showed for SAM 1 that encoder-only feature distillation is insufficient and that distilling through the prompt-conditioned decoder is what recovers accuracy.

Quantization and vocabulary caching. Post-training quantization of SAM has been studied down to 4 bits [27], building on integer inference [20]. YOLO-World [10] removes the text encoder for a fixed vocabulary by re-parameterising its outputs ("prompt-then-detect").

Architectural alternatives. Deep equilibrium models [1] replace a stack of layers with a single layer iterated to a fixed point, giving constant training memory in effective depth; MDEQ [2] extends this to multi-resolution vision features, and RevDEQ [28] makes the gradient exact through a reversible solver. Deformable DETR [53] replaces global cross-attention with attention to a few learned sampling points. Token merging [5, 32] reduces the

number of image tokens, and the SAM 3.1 object multiplex [29] shares computation across tracked objects in video.

2.6 Reproducible Evaluation

The reproducibility literature documents how evaluation choices manufacture results [11, 35, 37]. The remedies we adopt are standard: pre-registering the confirmatory analysis [33], freezing data splits, cluster bootstrap intervals over the independent unit [12], rank-based paired tests [17, 47] with multiplicity control [3, 18], and matched random controls.

2.7 Research Gap

Three gaps motivate this thesis. First, training-free domain adaptation for SAM 3 had been reported, including by our own preliminary work, without controls that could separate a real effect from an evaluation artefact. Second, the only released low-end SAM 3 had no published accuracy, latency or memory numbers, so it was unknown whether it is usable or why it is not. Third, no work had tested whether EdgeSAM's lesson—distil outputs, not features—transfers to SAM 3's text-prompted, presence-factored concept path.

Chapter 3

Preliminary Work

This chapter records what we proposed and reported in the preliminary phase of the project [22], in enough detail that the re-evaluation in later chapters can be compared against it. Numbers here are the preliminary ones; later chapters test them.

3.1 ConceptBank: Parameter-Free Domain Adaptation

3.1.1 Formulation

Let $\mathcal{S}$ be SAM 3's source domain (natural images) and $\mathcal{T}$ a target domain such as LoveDA aerial tiles. The goal was a calibration $C : \mathbb{R}^d \to \mathbb{R}^d$ that moves SAM 3's text embeddings toward $\mathcal{T}$ without changing model weights, intercepting the text features between the text encoder and the fusion encoder (Figure 2.1).

3.1.2 Prototype discovery

1. **Feature extraction.** Twenty support images from the target domain were passed through SAM 3's frozen feature pyramid to obtain 256-dimensional features on a 72×72 grid.
2. **Affinity and spectral clustering.** A cosine affinity matrix over the features of each image was clustered into K=3 groups from eigenvectors of its graph Laplacian.
3. **Prototypes.** Cluster means gave 60 raw prototypes, which were grouped by the dominant ground-truth class of each cluster and averaged into one prototype per foreground class.

3.1.3 Steering

Two ways of combining a retrieved prototype P with the text embedding were compared. *Blend* replaced the embedding with a mixture,

$$E_{\text{cal}} = \alpha\, E + (1 - \alpha)\, P, \qquad \alpha = \sigma(\tau \cos(P, \bar{E})), \tag{3.1}$$

Table 3.1: Preliminary results as reported [22] (LoveDA Urban, ten images).

Configuration	Reported mIoU	Δ	Reported interpretation
Baseline SAM 3	0.237	—	
Residual, $\lambda = 0.05$	0.256	+0.019	best; forest 0.000 → 0.053, agriculture +0.059
Residual, $\lambda = 0.10$ / 0.20	0.251 / 0.241	+0.014 / +0.004	larger steps harm strong classes
Pooled-only, $\lambda = 0.05$ / 0.20	0.252 / 0.226	+0.015 / −0.011	
Blend, $\tau \in \{0.1, 1, 10\}$	0.130–0.143	−0.095 to −0.107	"leaves the text manifold"
Residual + confidence gating	0.256	+0.019	no difference on this set

and *residual steering* nudged it,

$$E_{\text{cal}} = E + \lambda\,(P - \bar{E}), \qquad \lambda = 0.05, \tag{3.2}$$

where $\bar{E}$ is the pooled text embedding and P the prototype closest to it by cosine similarity. A third, token-weighted "pooled-only" variant was also swept.

3.1.4 Preliminary results

Evaluation used ten LoveDA Urban validation images and a separate support set of twenty. For each image and class, SAM 3 was prompted with the class name, and *the single highest-confidence mask* was compared with the class's ground-truth region at a confidence threshold of 0.3. Table 3.1 reproduces the reported results.

The interpretation offered was that blend fails because it moves the embedding off the manifold of text embeddings the decoder was trained on, whereas a small residual step stays close to it while adding "domain awareness".

3.2 Architectural Efficiency Proposals

The preliminary analysis counted transformer layers per component—32 in the ViT encoder and 6 each in the transformer encoder, fusion encoder and decoder—and concluded that deep stacks of globally attending layers were the bottleneck. It proposed four interventions (Table 3.2).

In practice, integrating DEQ into SAM 3's features ran into convergence problems, and a deformable-attention decoder was assembled and checked in a forward pass only, without training; accuracy and latency of the modified models were not measured.

Table 3.2: Preliminary efficiency proposals [22].

Method	Idea	Proposed target in SAM 3
DEQ [1]	Replace a layer stack by one layer iterated to a fixed point	Decoder stack or fusion encoder
MDEQ [2]	Joint equilibrium across feature resolutions	Bridge between backbone and mask decoder
RevDEQ [28]	Reversible fixed-point solver with exact gradients	Transformer-heavy components
Deformable DETR [53]	Attend to a few learned sampling points instead of the full map	Decoder cross-attention

3.3 Why a Re-Evaluation Was Necessary

Looking back at the preliminary phase with the questions of Section 1.3 in mind, several features of its evaluation meant that its conclusions could not be trusted in either direction:

- **Selection and reporting on the same images.** Step sizes, variants and the threshold were chosen on the same ten images the result was reported on.
- **A single, unexamined read-out rule.** Comparing one instance mask with a whole class region had never been checked on a case with a known answer.
- **No controls.** Nothing separated an effect of the *prototypes* from an effect of *moving the embedding at all*, and step sizes of different operator families were not matched.
- **Label use.** Prototypes were assigned to classes using ground-truth labels, which the "parameter-free, unsupervised" framing did not account for.
- **Cost by layer count.** Efficiency targets were chosen by counting layers, not by measuring parameters, operations or time, and no modified model was trained or timed.
- **Lost implementation.** The original code was not preserved.

The remainder of the thesis rebuilds everything from scratch under the framework of Chapter 4 and reports what survives.

Chapter 4

Research Methodology

This chapter describes the evaluation framework used in every later chapter. It was designed around the specific ways a small inference-time effect can be manufactured by evaluation choices, including the ways our preliminary evaluation was exposed to them (Section 3.3). Table 4.1 lists each failure mode and its countermeasure.

4.1 Datasets and Frozen Splits

- **LoveDA Urban and Rural** [46]: aerial 1024×1024 tiles at 0.3 m per pixel. Six foreground classes: building, road, water, barren, forest, agriculture (the preliminary study used five, omitting barren).
- **SUIM** [19]: underwater imagery (TEST split). Five classes: diver, plants, reef, fish, sea floor.
- **COCO val2017** [25]: natural photographs, in-distribution for SAM 3. Five classes: person, car, bird, dog, boat.

Splits were drawn once with a fixed seed and committed before any analysis (Table 4.2). The *support* tier is used only to build prototype banks, *dev* for all selection (thresholds, operators, step sizes, checkpoints) and *test* for reporting. Prompts are plain class names.

The **COCO confirmation set** was built after the 50-image tier proved too small to choose checkpoints and thresholds reliably (Section 8.5). It contains 500 val2017 images disjoint from every image used before, each with at least one non-crowd instance ($\geq 32 \times 32$ px) of the five classes, drawn round-robin across classes and split by seed into 100 dev and 400 test images. Its analysis plan was written before any model was evaluated on it.

4.2 From Instance Predictions to Semantic Masks

SAM 3 returns a set of scored instance masks per prompt, whereas the benchmarks provide one semantic mask per image. Three conversions are implemented:

`union` (primary). The prediction for class c is the union of every instance mask whose score is at least the threshold θ.

Table 4.1: Failure modes of an inference-time evaluation and the countermeasures used throughout.

Failure mode	Countermeasure
Read-out capped by construction	Three read-out protocols reported; a unit test pins union = 1.0 vs. top-1 = 0.1 on ten perfect instances
Aggregation fixed at collection	Records store raw TP/FP/FN per (image, class, operator, threshold); IoU derived at analysis time
Incomparable step sizes	Every embedding operator parameterised by its realised relative norm ρ
Selection on the reported images	Frozen splits; dev tier for choices; test tier read after pre-registration
No control for direction	19 matched-norm random directions
No control for prototypes	Shuffled-bank, constant-offset and wrong-domain operators
Effect is a threshold shift	Each arm compared at its own best threshold, with selection priced into the interval
Post-hoc narrative	Pre-registered endpoint and decision rule, committed before the test tier
Small evaluation set	Headline efficiency result confirmed on 400 new images with a plan fixed before running

Table 4.2: Frozen splits (images).

Dataset	support	dev	test	heldout	role
LoveDA Urban	80	24	50	523	aerial: viewpoint and scale shift
LoveDA Rural	80	24	50	838	aerial: sparse structures
SUIM	10	25	50	25	underwater: colour cast, turbidity
COCO val2017	10	15	50	15	in-distribution control; efficiency accuracy
COCO confirmation	—	100	400	—	efficiency confirmation (Section 8.6)

`argmax`. Classes compete per pixel on mask probability times score (robustness check).

`legacy_top1`. Only the single highest-scoring instance is kept—the preliminary protocol, used only to reproduce it.

Metric. The primary metric is *dataset-aggregated* mIoU: TP, FP and FN are summed over all images per class, IoU is computed once per class, and classes are averaged. Thresholds $\theta \in \{0.1, \ldots, 0.6\}$ are evaluated from one forward pass, since the decoded masks are probabilities and filtering by score is free. Two diagnostic quantities recur: the *median presence* probability on classes that are present, and the share of present classes with no surviving instance ("missed").

4.3 Matched-Norm Operators and Controls

Every embedding intervention is a map $E \mapsto E' = E + \Delta$ on one prompt's text features. To make operators with different formulas comparable, each is parameterised by its realised

relative step

$$\rho = \frac{\|\Delta\|_F}{\|E\|_F}, \tag{4.1}$$

computed over non-padding tokens (padding is never modified) and asserted to 10^{-4} on every record. Controls share the operator interface:

- **Random direction**: an isotropic Gaussian direction, seeded per (seed, class), rescaled to the same ρ. Nineteen seeds are used because a one-sided percentile test against n controls has a p-value floor of $1/(n+1)$, and 19 is the smallest n at which $p = 0.05$ is attainable.
- **Shuffled bank**: prototypes assigned to the wrong classes.
- **Constant offset**: one global offset for every class, with no retrieval.
- **Wrong domain**: prototypes from LoveDA Rural used on Urban.

4.4 Statistics

Rows from the same image share features and error structure, so all intervals come from a **paired cluster bootstrap over images**: each replicate resamples image identifiers with replacement, uses the same draw for both arms, and recomputes the aggregated mIoU difference; percentile 95% intervals are reported (2,000–10,000 resamples). Alongside, we report the Wilcoxon signed-rank test [47] on per-image differences, the Hodges–Lehmann estimate [17], win/loss/tie counts ($|\Delta\text{IoU}| \leq 0.005$ is a tie), and the minimum detectable effect at 80% power beside every null. Per-class tests use Benjamini–Hochberg control [3]. When each arm is compared at its own best threshold, the threshold is re-selected inside every bootstrap replicate so that the interval prices in the selection.

4.5 Pre-Registration

Before the LoveDA Urban test tier was read, we committed a pre-registration fixing the primary endpoint (residual ConceptBank at $\rho = 0.05$ versus no intervention, union protocol, dev-selected threshold, six classes), the decision rule (a confidence interval excluding zero, *and* beating all 19 random directions, *and* replication on Rural), secondary analyses and falsification criteria. Analyses not listed are labelled exploratory. The confirmation of the efficiency results (Section 8.6) had its own plan, written before evaluation.

4.6 Efficiency Measurement Protocol

- **Latency**: image encoding and per-concept grounding timed through exactly the calls the accuracy harness uses; median of repeats after a warm-up; CPU otherwise idle.

"Five-concept image" means one image encoding plus five per-concept passes.

- **Size**: serialized weights, or the exported file.
- **Memory**: resident set size sampled every 2 ms and attributed to stages (build, image encoder blocks, per-concept pass).
- **Parameters and FLOPs**: exact parameter counts from the built modules; analytic FLOPs from the configuration constants, cross-checked against measured latency.

4.7 Implementation and Hardware

The code base separates a model-facing bridge (the only module that imports SAM 3) from operators, metrics and statistics, which are covered by 134 model-free unit tests plus end-to-end tests against the real model; the efficiency code adds exactness tests (Appendix B). Accuracy experiments and training ran on an NVIDIA RTX 4060 (8 GB) in bfloat16. All latency and memory figures were measured on an AMD Ryzen 5 7600 CPU (6 cores, 12 threads, AVX-512 VNNI) with 14 GB of RAM, PyTorch 2.10. Every per-image result is stored as a record from which every table and figure is regenerated by script.

Chapter 5

Re-Evaluating ConceptBank

This chapter answers RQ1. We rebuilt ConceptBank from the preliminary report, design notes and shell history, reproduced its evaluation, corrected it, and tested whether the prototypes do any work.

5.1 The Rebuilt Method

The rebuild follows Section 3.1: FPN features of support images, spectral clustering with K=3, cluster-mean prototypes grouped by dominant ground-truth class, top-3 retrieval and residual steering (Equation (3.2)), intercepting the text features at the single point where SAM 3 writes them. Rebuilding surfaced two facts the preliminary framing did not state.

The method uses labels. Prototypes were assigned to classes by the *dominant ground-truth class* of each cluster, so the bank reads the target domain's pixel labels. ConceptBank as implemented is few-shot weakly supervised, not unsupervised.

The bank is thin where it matters. Thin classes are rarely dominant in any cluster and barely acquire prototypes (Table 5.1). Forest, the class credited with the preliminary headline gain, rests on three prototypes in the Urban bank; the Rural bank has no road or forest prototype at all.

A further implementation detail mattered: the eigenvector rule in the preliminary description (keep eigenvectors "at the smallest non-zero eigenvalues") fails to separate even trivially separable clusters; the rebuild uses the Ng–Jordan–Weiss form [31], pinned by a unit test.

5.2 The Read-Out Protocol Was Capped by Construction

The preliminary evaluation compared *the single highest-scoring instance mask* with *the entire ground-truth region of the class*. A LoveDA tile contains dozens of buildings, so even a perfect model is capped near the area of one instance divided by the area of all of them

Table 5.1: Prototype bank composition (20 support images, K=3 clusters each, labelled by dominant ground-truth class).

Bank	background	building	road	water	barren	forest	agriculture
Urban	15	11	4	9	2	3	16
Rural	35	1	0	9	1	0	14

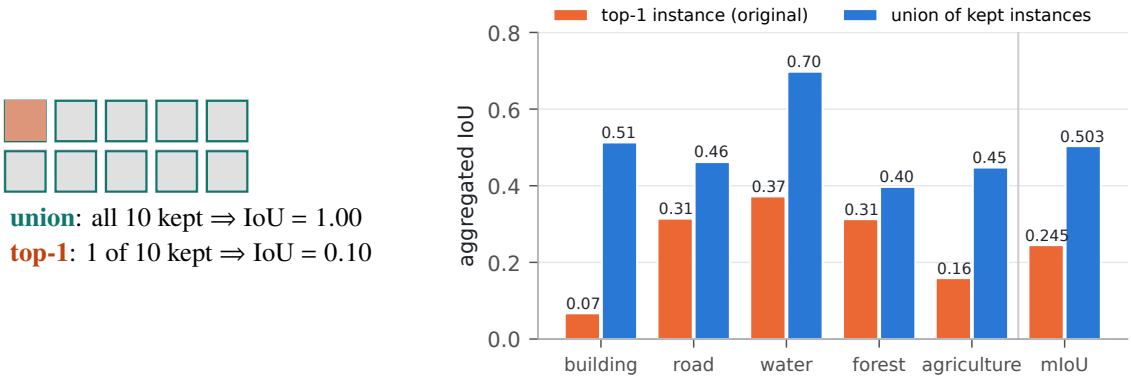


Figure 5.1: Left: why a top-1 read-out is capped on multi-instance classes. Right: the same model, weights and images scored by both protocols (LoveDA Urban, 50 held-out images, the preliminary five classes, threshold 0.3). Correcting the read-out doubles measured mIoU—about 14× the effect originally attributed to ConceptBank.

(Figure 5.1, left). A unit test pins this: for ten perfectly predicted, equally sized regions, `legacy_top1` scores 0.100 and `union` scores 1.000.

On 50 held-out LoveDA Urban images, with the preliminary five classes and threshold 0.3, SAM 3's aggregated mIoU is **0.2450** under the top-1 protocol and **0.5033** under the union protocol. The gap is concentrated where the construction predicts, in classes made of many separate instances: building rises from 0.07 to 0.51 and agriculture from 0.16 to 0.45, while contiguous classes change less (forest 0.31 to 0.40). Roughly half of "SAM 3 degrades severely on remote sensing" was measurement, not domain shift, and correcting the read-out moves measured performance about 14× more than ConceptBank was reported to (+0.258 versus +0.019).

One preliminary number is not explained by the protocol: forest was reported at 0.000 at baseline, but under the preliminary top-1 protocol our rebuild measures it at 0.31 on held-out images (0.30 on a ten-image subset). Without the original code, the source of that zero cannot be determined; the baseline it was compared against is not reproducible.

5.3 The Preliminary Gain Does Not Reproduce

We first reproduced the preliminary evaluation *under its own protocol* (top-1, threshold 0.3, five classes, similarity retrieval over labelled prototypes) on 50 held-out images (Table 5.2). The baseline reproduces (0.2450 versus 0.2373 reported). The improvement does not: the preliminary winning configuration gives +0.0001 with a 95% CI of $[-0.0026, +0.0030]$,

Table 5.2: Reproduction under the preliminary top-1 protocol (LoveDA Urban, 50 held-out images, threshold 0.3). ρ is the measured relative step. Baseline 0.2450; MDE 0.0090.

Configuration	Δ mIoU	95% CI	W / L / T	ρ
residual $\lambda = 0.05$ (preliminary best)	+0.0001	[-0.0026, +0.0030]	2 / 2 / 46	0.034
residual $\lambda = 0.10$	−0.0061	[-0.0173, +0.0062]	6 / 9 / 35	0.068
residual $\lambda = 0.20$	−0.0105	[-0.0279, +0.0081]	10 / 13 / 27	0.135
blend $\tau = 10$	−0.1320	[-0.1755, -0.0798]	2 / 39 / 9	0.510
blend $\tau = 1$	−0.1388	[-0.1800, -0.0892]	1 / 44 / 5	0.501

and 46 of 50 images are unchanged to within 0.005 IoU. The minimum detectable effect is 0.0090, less than half the claimed effect, so this is a well-powered null.

The blend collapse does reproduce—but the last column shows why the preliminary interpretation does not follow. λ and τ are not commensurable: at the preliminary settings blend moved the embedding by $\rho \approx 0.51$ and residual by $\rho = 0.034$, a 15× larger step. At matched ρ on the dev tier the two families converge as $\rho \to 0$ (−0.0009 versus −0.0007 at $\rho = 0.005$) and diverge only at large steps (−0.049 versus +0.009 at $\rho = 0.2$). Residual steering translates the token cloud; blend translates it *and* contracts it toward P.

5.4 Why It Cannot Work: The Geometry of Text and Image Features

We measured the quantities that retrieval and steering depend on, using the 60 Urban prototypes and the pooled text embeddings of the class prompts:

prototype-to-text norm ratio $\|P\|/\|\bar{E}\|$	0.06
$\cos(\bar{E}, P)$: mean (sd), range	−0.041 (0.083), [−0.228, +0.139]
retrieval top-1 accuracy (uniform chance)	0.183 (0.167)
label-permutation test (10,000 permutations)	$p = 0.63$

Retrieval is at chance. Cosine similarity between a pooled text embedding (a linear resizer output) and an FPN prototype (a convolutional neck output) does not recover the prototype's class. The two spaces are nearly orthogonal and differ in scale by 17×—an instance of the cone effect and modality gap [24]. SAM 3 only reconciles them inside the fusion encoder's cross-attention, which uses its own projections for each.

The steering direction does not depend on the prototype.

Proposition 5.1. *Let $\delta = P - \bar{E}$ with $\|P\| = \kappa\|\bar{E}\|$ and $\kappa < 1$. Then the angle between δ and $-\bar{E}$ is at most* $\arcsin \kappa$*, whatever the direction of P.*

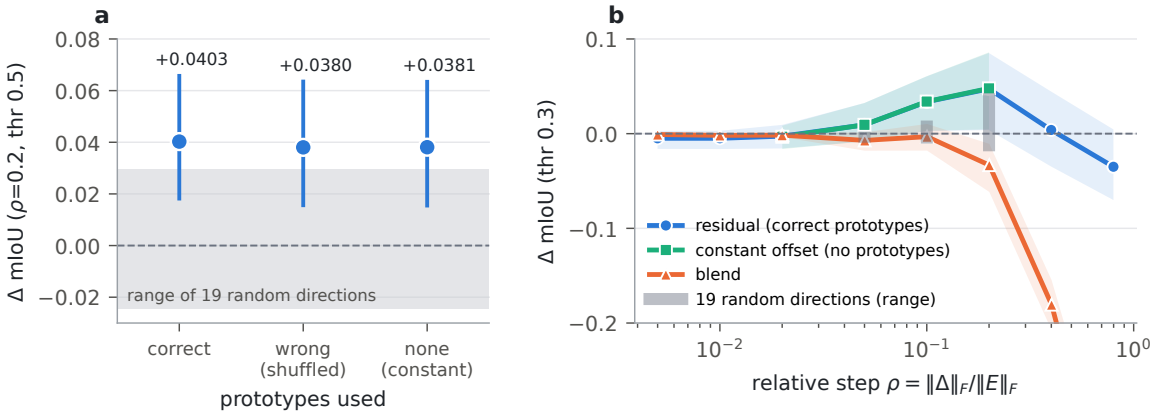


Figure 5.2: Pre-registered test tier (LoveDA Urban, N=50). Left: at $\rho = 0.2$, correct, deliberately wrong and absent prototypes give the same gain (95% CIs). Right: response to step size for residual steering against matched-norm random directions (band: range over seeds). An effect exists, but it is a direction effect along $-\bar{E}$, not a prototype effect.

Table 5.3: The decisive control (LoveDA Urban test, N=50, $\rho = 0.2$, threshold 0.5, baseline 0.3323).

Operator	Uses	Δ mIoU	95% CI	W / L / T	Wilcoxon p
residual	correct prototypes	+0.0403	[+0.0175, +0.0671]	30 / 10 / 10	3×10^{-5}
shuffled	wrong prototypes	+0.0380	[+0.0152, +0.0649]	29 / 10 / 11	1×10^{-4}
constant offset	nothing	+0.0381	[+0.0153, +0.0651]	29 / 10 / 11	1×10^{-4}

(Proof in Appendix A.) With $\kappa = 0.06$ every ConceptBank step lies within 3.4° of $-\bar{E}$. Equation (3.2) is therefore $E - \lambda\bar{E} + \lambda P$: a *partial mean-centering of the prompt's own embedding*, with the prototype contributing a 6% perturbation. Whatever the method does, it cannot be injecting target-domain appearance.

5.5 The Controls Confirm It

Pre-registered primary endpoint: not supported. Residual ConceptBank at $\rho = 0.05$, at the dev-selected threshold 0.5, gives +0.0003 [−0.0111, +0.0088] (W/L/T 12/5/33, MDE 0.012) and does not beat the 19 matched random directions (mean +0.0010, maximum +0.0147, $p = 0.60$). It fails the first two criteria of the pre-registered decision rule.

A larger step does something—and the prototypes are irrelevant to it. Exploring step size (Figure 5.2, right), residual steering improves with ρ, beats every random direction at $\rho = 0.05$ and 0.1 ($p = 0.05$, the floor), peaks near $\rho \approx 0.2$ and collapses by $\rho = 0.8$, when the embedding stops resembling text. The decisive comparison is at $\rho = 0.2$ (Table 5.3): correct prototypes, prototypes of the *wrong* class, and *no prototypes at all* (one constant offset) produce the same gain within 0.002, each with an interval excluding zero. Synonym prompts, the cheapest alternative, were worse than plain class names (−0.026 and −0.188 on dev).

Rural. On LoveDA Rural at threshold 0.5 the equivalence replicates (residual +0.0056, constant offset +0.0058, shuffled +0.0053; random directions −0.0026 to +0.0010). But the Rural baseline peaks at threshold 0.2 (0.3719), and compared at their own best thresholds residual and constant offset give −0.0092 and −0.0078: the fixed-threshold Rural gain is a threshold effect.

5.6 Summary

The whole apparatus—support images, feature extraction, spectral clustering, the prototype bank and retrieval—does no work that a fixed offset does not do. The preliminary gain was not reproducible, its evaluation understated SAM 3 by half, and the blend-versus-residual comparison confounded direction with step size. What remains is a genuine direction effect along $-\bar{E}$ that needs no target-domain data. Chapter 6 characterises it.

Chapter 6

A Data-Free Recalibration and Its Mechanism

Chapter 5 left a direction effect along $-\bar{E}$ that needs no prototypes. This chapter answers RQ2: it isolates that effect as an operator using no data at all, characterises it through a two-parameter decomposition of the text features, measures how large it really is once every selection step is accounted for, and identifies the mechanism.

6.1 Mean-Shrink: ConceptBank Without Prototypes

Letting the prototype vanish in Equation (3.2) leaves

$$E'_t = E_t - s\,\bar{E} \qquad \text{for every non-padding token } t, \tag{6.1}$$

with s chosen so that the realised step is ρ. This *mean-shrink* operator needs no support images, clustering, bank or target-domain data. At matched $\rho = 0.2$ on the Urban test tier it is indistinguishable from the bank-based constant offset (threshold 0.3: +0.0474 versus +0.0478), so nothing upstream of it in ConceptBank carries information. It is deployable in a few lines at the interception point of Figure 2.1.

6.2 The Gist/Detail Decomposition

Definition 6.1. For a prompt with non-padding tokens $\mathcal{V}$, write $E_t = \bar{E} + R_t$, where $\bar{E} = \frac{1}{|\mathcal{V}|}\sum_{t\in\mathcal{V}} E_t$ is the *gist* (shared by the whole phrase) and R_t the per-token *detail*. The gist/detail operator is $E'_t = \alpha\,\bar{E} + \beta\,R_t$, with ratio $r = \alpha/\beta$; its *norm-controlled* variant rescales E' back to $\|E\|_F$.

Proposition 6.1. *(i) Gist and detail are orthogonal in Frobenius inner product, so* $\|E\|_F^2 = |\mathcal{V}|\,\|\bar{E}\|^2 + \|R\|_F^2$*; define the gist fraction* $g = \sqrt{|\mathcal{V}|}\,\|\bar{E}\|/\|E\|_F$*. (ii) Mean-shrink at step* ρ *equals the gist/detail operator with* $\alpha = 1 - \rho/g$, $\beta = 1$*. (iii) The norm-controlled operator depends on* (α, β) *only through* r*.*

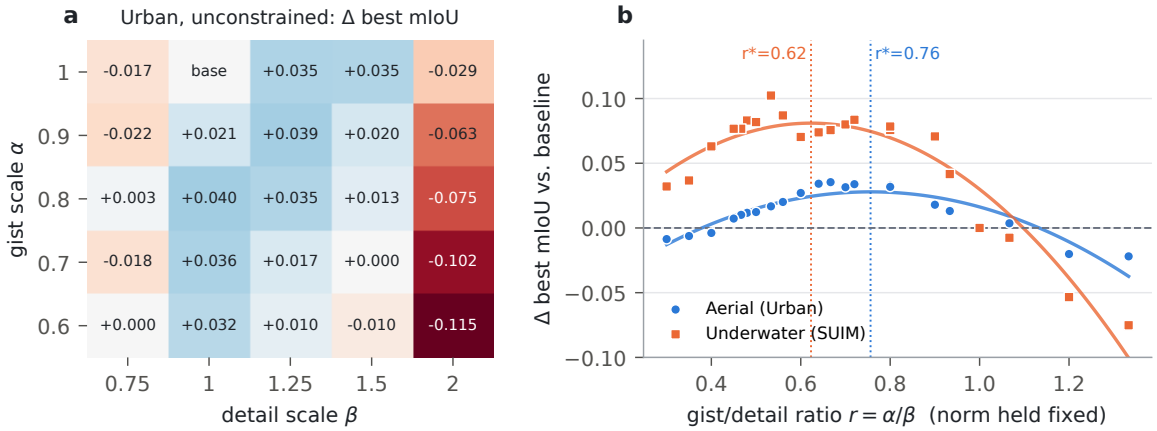


Figure 6.1: The gist/detail plane (exploratory). (a) Unconstrained sweep on Urban: change in best-threshold mIoU versus the baseline's best. Gains appear both by shrinking the gist (β=1) and by amplifying detail (α=1). (b) Norm-controlled sweeps: accuracy is an inverted-U function of r in both domains, with optima $r^\star \approx 0.76$ (aerial) and 0.62 (underwater).

(Proof in Appendix A.) SAM 3's prompts are gist-dominated: measured g is 0.63–0.69 for the LoveDA classes, 0.62–0.76 for SUIM and 0.63–0.70 for COCO. Two consequences follow. Mean-shrink is one line ($\beta = 1$) of a plane that also contains the opposite operation, *amplifying detail while removing nothing* ($\alpha = 1, \beta > 1$). And under norm control the plane collapses to the single parameter r *by construction*, so agreement between cells that share a ratio is not evidence of anything.

6.3 Exploring the Plane

We swept $\alpha \in \{0.6, \ldots, 1.0\}$ and $\beta \in \{0.75, 1, 1.25, 1.5, 2\}$ on the Urban test tier with and without norm control, and repeated the norm-controlled sweep on SUIM (Figure 6.1). These sweeps are *exploratory*: every cell and threshold was examined on test images, so their maxima are optimistic and are not used as effect estimates.

Three observations survive the caveat. *First*, removal is not the operative ingredient: amplifying detail with the gist untouched (α=1, β=1.25: +0.035) is as effective as shrinking the gist (α=0.8, β=1: +0.040). This distinguishes the effect from All-but-the-Top [30], which removes a corpus-level common component; here the variable is the within-prompt balance of gist and detail. *Second*, in the unconstrained plane the ratio is the primary variable but scale is not irrelevant: cells that reach $r = 0.8$ by shrinking both components score far lower than those that do not. *Third*, the preferred ratio is similar across physically unrelated shifts (nadir viewpoint versus colour cast): $r^\star \approx 0.76$ aerial and 0.62 underwater, both below SAM 3's default balance $r = 1$.

Table 6.1: The recalibration's effect under increasingly strict handling of selection (N=50 per test tier). Level 1: thresholds selected on the Urban dev tier. Level 2: each arm at its own best test threshold, re-selected inside every bootstrap replicate; p_{rand} is the percentile test against 19 random directions treated the same way. Level 3: operator and threshold rule fixed in advance on Urban and applied unchanged to SUIM.

Level	Domain, operator	Δ mIoU	95% CI	P(Δ≤0)	p_{rand}	W / L / T
1: dev-selected thresholds	Urban, constant offset ρ=0.1	+0.023	[+0.005, +0.039]	—	0.05	30 / 9 / 11
	Urban, residual ρ=0.1	+0.022	[+0.005, +0.039]	—	0.05	31 / 9 / 10
2: best vs. best, selection-aware	Urban, shrink gist (0.8, 1.0)	+0.040	[-0.012, +0.057]	0.06	0.05	
	Urban, amplify detail (1.0, 1.25)	+0.035	[-0.012, +0.058]	0.09	0.10	
	SUIM, amplify detail (1.0, 1.25)	+0.090	[+0.023, +0.118]	0.002	0.05	
	COCO, amplify detail (1.0, 1.25)	−0.002	[-0.018, +0.006]	0.89	0.35	
	Rural, constant offset ρ=0.05	−0.008	[-0.022, +0.008]	0.78	0.83	
3: out-of-sample	SUIM, amplify detail (1.0, 1.25)	+0.090	[+0.024, +0.161]	—	0.05	22 / 6 / 21

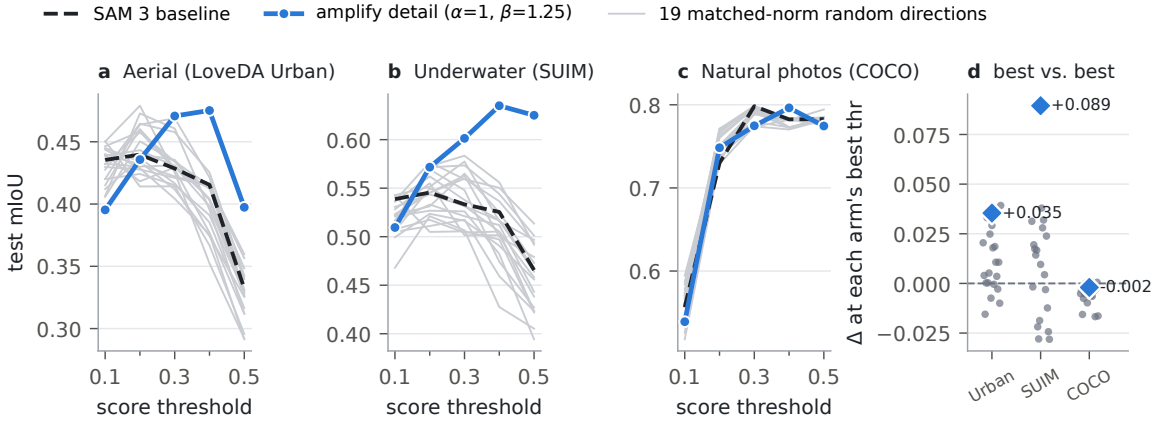


Figure 6.2: Full threshold curves for the amplify-detail operator chosen on Urban: baseline, operator and 19 matched-norm random directions on each domain, and each arm at its own best threshold. Underwater the operator clears the entire random envelope; on aerial imagery its best-versus-best gain is within reach of the best random direction; on COCO nothing changes.

6.4 How Large Is the Effect?

A small effect in a thresholded pipeline can be inflated in two ways, both of which our own intermediate analyses fell into: comparing arms at a threshold that handicaps the baseline, and selecting operator and threshold on the reporting images. We therefore report the effect at three levels of stringency (Table 6.1).

Aerial imagery: small and fragile. With thresholds chosen on dev, the Urban gain is about +0.02 (+0.023 [+0.005, +0.039], Wilcoxon $p = 0.004$, beating all 19 random directions). Letting each arm use its best test threshold raises point estimates to +0.035–+0.040, but once that selection is priced in every Urban interval includes zero, and the best random direction reaches +0.039 (Figure 6.2). The effect on aerial imagery is real in direction and consistent across operators, but its reliable size is about +0.02 mIoU; on Rural there is no

Table 6.2: Per-class IoU under the out-of-sample rule (baseline at threshold 0.2, operator at 0.4). n: test images containing the class.

SUIM, amplify detail					Urban, amplify detail				
class	n	base	op.	Δ	class	n	base	op.	Δ
diver	15	0.721	0.898	+0.177	building	44	0.597	0.575	−0.022
fish	36	0.545	0.685	+0.140	road	44	0.446	0.545	+0.099
plants	10	0.189	0.240	+0.051	water	46	0.405	0.418	+0.013
reef	22	0.541	0.581	+0.040	barren	25	0.275	0.373	+0.098
sea floor	27	0.731	0.771	+0.040	forest	44	0.459	0.482	+0.023
					agriculture	41	0.456	0.458	+0.002

gain once thresholds are handled fairly.

Underwater imagery: large, and out of sample. The amplify-detail operator was fixed on Urban before any SUIM image was evaluated. Applied unchanged it gives 0.5455 → 0.6349, +0.090 [+0.024, +0.161], Wilcoxon $p = 0.002$, and exceeds all 19 matched random directions (best random +0.038). Every class improves (Table 6.2), most of all divers and fish.

Natural photographs: nothing. On COCO the operator is neutral at each arm's best threshold (−0.002 [−0.018, +0.006]).

Context. Supervised parameter-efficient fine-tuning on LoveDA reports 52–53% mIoU [4, 9, 16], against about 44% for zero-shot SAM 3 under our corrected protocol. The numbers are not like-for-like, but they bound what inference-time recalibration should deliver on aerial imagery: a reliable +0.02, obtained with neither labels nor training.

6.5 Mechanism: A Presence Veto

Rescaling a prompt's gist could help through localisation (better $\sigma(\ell_q)$ or better masks) or through the prompt-level presence estimate $\sigma(\ell_p)$ (Equation (2.2)). We recorded both factors for every detection and classified every *present* (image, class) pair: *missed* if no instance clears the threshold, *rescued* if the baseline misses it and the operator does not, *lost* in the opposite case; an absent pair becomes a *false alarm* if only the operator keeps an instance.

Table 6.3 and Figure 6.3 give a consistent picture:

1. **The gain is rescued detections**: 123% of the net gain on aerial imagery (masks on classes both arms found get slightly worse) and 87% underwater, at every threshold from 0.2 to 0.5.

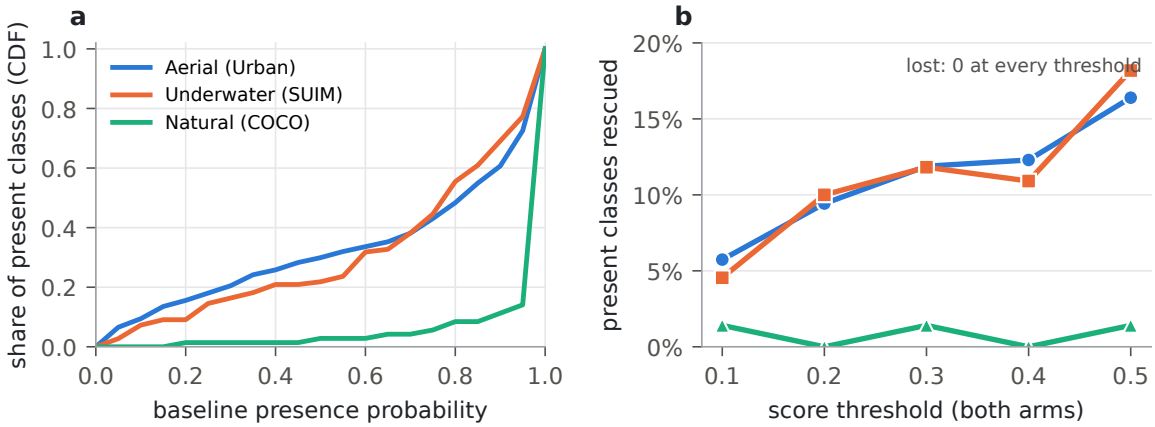


Figure 6.3: The presence veto. (a) SAM 3's baseline presence on classes that *are* present: saturated on COCO; low for a large share of real classes on aerial and underwater imagery. (b) Present classes rescued at each threshold by the amplify-detail operator; none is ever lost. (c) Mean change in the two score factors.

Table 6.3: What changes when the gist/detail balance is shifted (amplify detail; both arms at the same threshold: 0.3 aerial, 0.4 underwater and COCO).

	Aerial (Urban)	Underwater (SUIM)	COCO
median baseline presence, present classes	0.809	0.783	0.988
present classes missed by the baseline	66 / 244 (27%)	24 / 110 (22%)	2 / 71 (3%)
rescued by the operator	29	12	0
lost by the operator	0	0	0
share of net IoU gain from rescued classes	123%	87%	—
false alarms added (absent classes)	9	13	3
mean Δ presence probability	+0.118	+0.127	+0.019
mean Δ best query probability	+0.010	+0.005	+0.005

2. **The operator acts on presence, not localisation**: presence rises by 0.12–0.13, 12–25 times more than the best query probability. The objects were already localised; a pessimistic prompt-level veto suppressed them.
3. **It never removes a correct detection**: across all thresholds and domains, no present class found by the baseline is lost.
4. **The cost is false alarms**, because presence also rises for absent concepts.

This explains why each arm's best threshold moves up (raising presence inflates every score of a prompt), and why the effect is not *merely* a threshold shift (presence is raised per prompt, most where it was low). An earlier hypothesis, that rebalancing "sharpens cross-attention over text tokens", predicts better masks on already-found objects, which we do not observe; we withdraw it.

Table 6.4: Presence gating with the gate fixed a priori (0.9), versus applying the operator always.

	always on	presence-gated	gate fired on
Aerial	+0.0354	+0.0173	67% of prompts
Underwater	+0.0895	+0.0894	85%
COCO	−0.0020	−0.0014	74%
mean	+0.0410	+0.0351	

6.6 A Confirmed Prediction: No Veto, No Gain

If the mechanism is a veto applied to unfamiliar imagery, then on familiar imagery, where presence is already confident, there should be nothing to rescue and no gain. COCO tests this, and the prediction holds (Table 6.3, right): median presence on present classes is 0.988, only 2 of 71 present classes are missed, the operator rescues none of them, and presence barely moves. Every operator lands at or slightly below baseline on COCO (amplify detail −0.0020, shrink gist −0.0105, mean-shrink −0.0109). The veto the operator repairs is active out of distribution and inactive in distribution.

6.7 A Refinement That Did Not Survive: Presence Gating

The mechanism suggests applying the operator only when SAM 3 doubts a concept, i.e. when baseline presence is below a gate. Searched over the gate and both arms' thresholds on the test tiers, gating looked strictly better (mean +0.049). That was selection on the test set. With the gate fixed a priori at 0.9 and pre-assigned thresholds, gating is worse than always-on (Table 6.4).

The failure is structural: on COCO the gate fires on 74% of prompts because presence is *correctly* low whenever a concept is genuinely absent. A presence gate cannot distinguish "present but vetoed" from "truly absent". This result reproduced, inside our own analysis, the error behind the preliminary ConceptBank claim—free parameters tuned on the reporting data—and was caught only by re-running with every choice fixed in advance.

6.8 Summary

The only part of ConceptBank that works is a data-free rebalancing of a prompt's gist against its detail. It lifts a pessimistic presence estimate and rescues objects SAM 3 had already localised: by +0.090 out of sample underwater, by a reliable +0.02 on aerial imagery, and not at all on photographs, where there is no veto to lift.

Chapter 7

Running SAM 3 on Low-End Hardware

The second objective of the preliminary phase was efficiency, proposed as deep equilibrium models [1, 2, 28] and deformable attention [53] for SAM 3's transformer components (Section 3.2). This chapter answers RQ3 and RQ4. It first measures where SAM 3's cost actually is and re-assesses the preliminary proposals against that measurement; it then measures the only released low-end SAM 3, locates the component that loses its accuracy, and evaluates two interventions: INT8 export with a cached vocabulary, and sharing per-concept computation. Unless stated otherwise, accuracy is COCO test (N=50, five classes, union protocol) and latency and memory are measured on the CPU of Section 4.6.

7.1 Where the Cost Is

Table 7.1 ranks SAM 3's components by parameters and FLOPs rather than by layer count, which is how the preliminary proposal chose its targets. Three facts follow.

1. **The image encoder dominates computation.** The ViT and neck are 93% of the FLOPs of one image with one prompt and 11.5 s of CPU time.
2. **Attention is not the operative constraint.** All attention score and value products together are 15.3% of the FLOPs; the other 84.7% is dense linear algebra and convolution.
3. **The text encoder is 42% of the weights and almost none of the work.** Its output depends only on the prompt string, so for a fixed vocabulary it can be replaced by stored outputs.

7.2 Re-Assessing the Preliminary Proposals

Deformable attention. Deformable attention would replace the decoder's image cross-attention, 6.4 GFLOPs or 0.10% of the image path; deleting the entire decoder saves 0.23%, which by Amdahl's law caps the speed-up at 1.002×. The decoder does have a real cost that FLOPs do not show: its box-relative positional bias is a dense $200 \times 5184 \times 8$ tensor per layer, and passing a dense mask to the attention kernel disables the fused backend, so

Table 7.1: Cost census of SAM 3. Parameters are exact; FLOPs are analytic for one 1008^2 image and one prompt, derived from the configuration constants and consistent with measured latency. CPU times are measured (float32, 12 threads).

Component	Params (M)	GFLOPs	FLOP share	Runs
ViT image encoder (32 blocks, d=1024)	446.2	5398.6	86.8%	once per image
FPN neck	7.8	410.6	6.6%	once per image
Text encoder	353.7	—	—	once per concept
Fusion encoder (6 layers, 5184 tokens)	9.5	255.9	4.1%	once per concept
Pixel decoder + mask head	2.3	141.7	2.3%	once per concept
DETR decoder (6 layers, 200 queries)	11.6	14.6	0.2%	once per concept
of which image cross-attention	—	6.4	0.1%	
Geometry encoder, scorer	9.4	—	—	once per concept
Total	840.5	6221.3		
Measured, CPU float32: image encode 11.48 s; per concept 1.07 s (of which text encoder 98 ms).				

the attention matrix is materialised in all six layers. Removing that is worth a few percent of the per-concept path, under 1% end to end. Moreover, deformable attention introduces sampling-offset and attention-weight projections with no pretrained values, so it cannot be evaluated in the forward-pass-only setting the preliminary phase used: it requires retraining the decoder.

DEQ, MDEQ and RevDEQ. The principal benefit of deep equilibrium models is memory that is constant in effective depth *during training*. At inference a fixed-point solve typically needs more evaluations of the block than the six explicit layers it replaces, so latency does not fall. The proposed targets (decoder, fusion encoder) hold 2.5% of the parameters and at most 4.1% of the FLOPs, and RevDEQ's contribution, exact gradients, has no effect without a backward pass. The convergence problems encountered in the preliminary phase are consistent with this: a pretrained explicit stack is not a contraction, and making it one requires training.

Conclusion for RQ3. The preliminary proposals are implementable but target components worth 0.1–4% of compute, and all require retraining. They were not pursued. What matters on low-end hardware is the image encoder (time), the text encoder (weights) and, as shown next, the per-concept path once the image encoder is made small.

7.3 Baseline: EfficientSAM3, Measured

EfficientSAM3 [49] replaces SAM 3's image encoder with a small backbone and its text encoder with MobileCLIP, keeping SAM 3's neck, fusion encoder, decoder and heads (we verified these are byte-identical to the released SAM 3 checkpoint). Its report publishes

Table 7.2: EfficientSAM3 students versus SAM 3 on COCO test (N=50). Latency: CPU, 12 threads, image encode plus five concepts. Sizes include the text encoder.

Model	Image backbone	mIoU @0.3	mIoU (best θ)	Image encode	Per concept	5-concept image	Size
SAM 3	ViT (446 M)	0.799	0.799 (0.3)	11.48 s	1.07 s	16.8 s	3372 MB
EV-M	EfficientViT-B1	0.332	0.380 (0.5)	0.87 s	1.13 s	6.5 s	390 MB
RV-M	RepViT-M1.1	0.323	0.329 (0.6)	—	—	—	—
TV-M	TinyViT-11M	0.274	0.427 (0.6)	1.26 s	1.14 s	7.0 s	415 MB

Table 7.3: Is the student's loss specific to domain shift? Test tiers (N=50 each), dev-selected thresholds, paired 95% CIs.

Domain	SAM 3	EV-M	Δ [95% CI]	Relative
COCO (in-distribution)	0.798	0.332	−0.467 [-0.559, -0.375]	−58%
LoveDA Urban	0.440	0.256	−0.184 [-0.234, -0.132]	−42%
LoveDA Rural	0.341	0.126	−0.215 [-0.257, -0.171]	−63%
SUIM	0.533	0.292	−0.241 [-0.347, -0.141]	−45%

no accuracy or latency. We evaluated the released checkpoints through our harness. As a check on the harness, SAM 3 run through EfficientSAM3's code matches the reference implementation ($|\Delta| \leq 0.005$ mIoU at threshold 0.3 on all four datasets), and CPU float32 and GPU bfloat16 agree within 0.015.

Table 7.2 shows two things. First, EfficientSAM3 is 13× faster at image encoding but *not* at concept processing: at five concepts, 87% of its time is the unchanged per-concept path. Second, it loses more than half of SAM 3's accuracy on COCO. This is not a robustness effect (Table 7.3): the loss is *largest* in distribution (−58% relative on COCO) and substantial on every domain. The failure signature differs by domain. On COCO presence is intact (median 0.98) but false positives flood (1.44–2.05 per true positive, against 0.14 for SAM 3); on aerial and underwater imagery presence collapses (median 0.51 Urban, 0.24 Rural, 0.20 SUIM, against 0.81 and 0.78 for SAM 3 on Urban and SUIM).

7.4 Which Distilled Component Loses the Accuracy?

EfficientSAM3 changes two components at once. We separate them with a 2×2 design built from released checkpoints: SAM3-LiteText [50] (small text encoder, SAM 3 image encoder) and EfficientSAM3's Stage-1 checkpoint (small image encoder trained only to imitate SAM 3's image features, SAM 3 text encoder).

Table 7.4 is unambiguous. Text distillation is nearly free except underwater, and for a fixed vocabulary it is unnecessary (Section 7.5). **The image encoder is where the accuracy goes, and it goes through presence**: the feature-imitation student loads cleanly (all 983 keys, no shape mismatch) yet drives SAM 3's presence estimate to 0.00–0.04 *even on COCO*,

Table 7.4: Component attribution: Δ mIoU versus SAM 3 at threshold 0.3, paired 95% CIs, N=50 per domain. Median presence on present classes is shown for the vision-only student.

Distilled component	COCO	Urban	Rural	SUIM
Text only (LiteText S0)	−0.026 [−0.059, 0.000]	+0.007 [−0.022, 0.038]	−0.015 [−0.039, 0.009]	−0.174 [−0.251, −0.105]
Vision only (Stage 1)	−0.748 [−0.808, −0.678]	−0.419 [−0.474, −0.369]	−0.341 [−0.375, −0.303]	−0.533 [−0.609, −0.456]
presence (median)	(0.04)	(0.01)	(0.00)	(0.00)
Both, fine-tuned (EV-M)	−0.467 [−0.558, −0.374]	−0.180 [−0.255, −0.117]	−0.284 [−0.330, −0.242]	−0.375 [−0.473, −0.273]

so almost nothing survives the veto of Equation (2.2). Its features are close enough to look plausible and far enough for the presence head to reject every concept. EfficientSAM3's later supervised fine-tuning stage recovers only part of the loss. This diagnosis motivates Chapter 8.

7.5 INT8 SAM 3 with a Cached Vocabulary

7.5.1 Method

Three changes, none of which requires data or training:

1. **Dynamic INT8 quantization.** Every linear layer receives per-channel INT8 weights, and activations are quantized at run time [20]. This covers the ViT's attention projections and MLPs (most of the compute) and the feed-forward layers of the fusion encoder and decoder.
2. **Fixed-vocabulary text cache.** The text encoder's three output tensors depend only on the prompt string. For a known class list they are computed once in float32 and stored, and the 1.4 GB text encoder is removed, following the prompt-then-detect idea of YOLO-World [10]. This is exact, not an approximation; a name outside the vocabulary raises an error.
3. **Export once, load directly.** The quantized model with its cache is serialised once and loaded straight into its final form.

7.5.2 Where the memory went

Table 7.5 shows that *the memory peak came from building, not from running*: constructing INT8 SAM 3 holds the float checkpoint, the float model and the quantized copy at once. Exporting once removes the peak. A second expectation was also wrong: the ViT's four global-attention blocks, whose full attention matrix over 72×72 tokens would need about 1.7 GB, do not raise memory above the windowed blocks on CPU, because the attention

Table 7.5: Resident memory by stage (MB), sampled every 2 ms, 12 threads.

Stage	INT8 built in memory	INT8 exported, loaded
Python and PyTorch, before any model	712	712
Build or load (peak)	**7835** (12.7 s)	1665 (0.9 s)
After build or load	2636	1665
Image encoder (peak)	3160	2284
One concept (peak)	3449	2641
Process peak	7471	**2525**

Table 7.6: INT8 SAM 3 on CPU (12 threads), COCO test (N=50). The exported model is bit-identical to the in-memory quantized model (maximum absolute difference 0.0 on logits, boxes, masks, presence and text features).

	float32	INT8, built in memory	INT8, exported
Image encode / per concept	11.48 s / 1.07 s	6.12 s / 1.01 s	same computation
Five-concept image	16.8 s	11.2 s (1.5×)	same computation
File size	3372 MB	1380 MB	**657 MB** (5.1×)
Peak RSS	7.6 GB	7.3 GB	**2.5 GB**
mIoU @0.3	0.799	0.774	0.774
Δ vs. float32 [95% CI]	—	−0.025 [-0.054, -0.001]	−0.025 [-0.054, -0.001]

kernel never materialises the matrix. Of the remaining 2.5 GB, about 0.7 GB is the PyTorch runtime itself.

7.5.3 Speed, size and accuracy

Table 7.6 summarises the result on the 50-image tier. Three observations matter. *The small loss is in the ViT and does not hold up*: quantizing only the ViT gives the same 0.774, almost all of the difference is one class (dog, 0.70 → 0.61), and on the 400 unseen confirmation images (Section 8.6) INT8 and float SAM 3 are indistinguishable: 0.810 versus 0.809 at dev-selected thresholds (+0.001 [-0.004, +0.006]), and +0.000 [-0.004, +0.005] at threshold 0.3. Quantizing the text encoder changes its output by about 13% but leaves mIoU unchanged; the export keeps float text features regardless. *INT8 does not accelerate the per-concept path* (1.07 → 1.01 s), which is dominated by attention, convolutions and normalisation rather than large linear layers; at five concepts it is 45% of INT8 SAM 3's time. *File size falls* 5.1× because the quantized weights are stored once and the text encoder is gone.

7.6 Can the Per-Concept Computation Be Shared?

Once the image encoder is fast, the per-concept path dominates (87% of EfficientSAM3's time at five concepts), and about half of it is the six-layer fusion encoder, re-run for every concept. SAM 3.1 already shares computation across tracked objects in video [29]; nothing

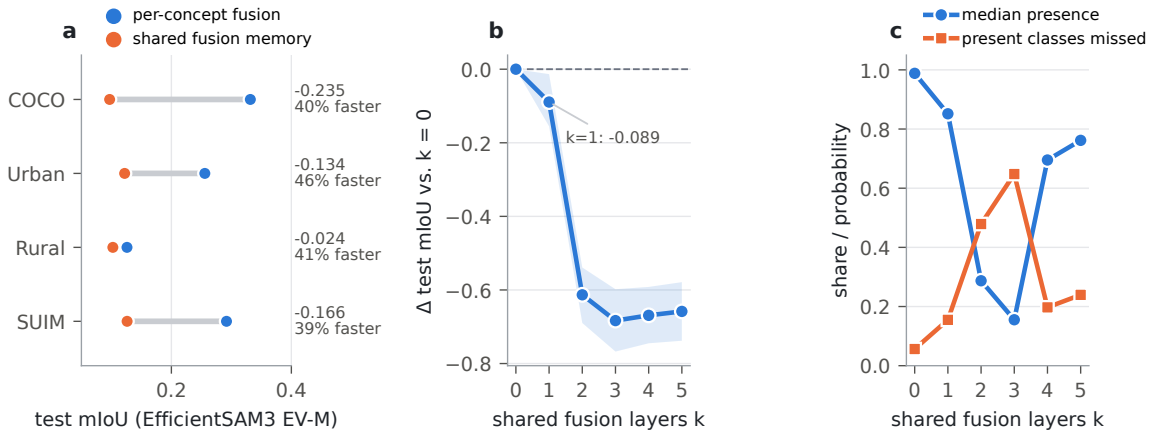


Figure 7.1: Sharing the fusion encoder across concepts. (a) Full sharing in EfficientSAM3 EV-M: 39–46% faster per image, but −0.02 to −0.24 mIoU (test tiers). (b) Partial sharing of the first k of six fusion layers in SAM 3 (COCO test at dev-selected thresholds, 95% CI); the pre-registered rule selects k=0. (c) Median presence and the share of present classes missed: once concept text is withheld from the fusion encoder, presence stops tracking the concept.

Table 7.7: Partial sharing of the first k fusion layers in SAM 3 (COCO; dev N=15, test N=50). Δ is relative to no sharing at its dev-selected threshold.

k shared	dev mIoU (thr)	test mIoU	Δ [95% CI]	median presence	present missed
0	0.727 (0.6)	0.790	—	0.99	6%
1	0.711 (0.4)	0.701	−0.089 [-0.155, -0.013]	0.85	15%
2	0.296 (0.2)	0.177	−0.613 [-0.690, -0.539]	0.29	48%
3	0.124 (0.2)	0.107	−0.683 [-0.767, -0.598]	0.16	65%
4	0.104 (0.3)	0.121	−0.669 [-0.745, -0.592]	0.70	20%
5	0.145 (0.4)	0.132	−0.658 [-0.738, -0.579]	0.76	24%

does so across several concepts in one image. We tested two designs:

- **Full sharing**: encode the image once against an empty prompt and give that memory to every concept; the concept text still drives the decoder, scorer and mask head.
- **Partial sharing**: run only the first k fusion layers once per image and the remaining $6-k$ per concept. The selection rule, fixed before running, is the largest k whose dev mIoU is within 0.01 of no sharing.

Both implementations reproduce SAM 3's own grounding bit-exactly when fed each concept's own memory (Appendix B), so any accuracy change is caused by sharing, not by plumbing.

Figure 7.1 and Table 7.7 show the outcome. Full sharing cuts forward time by 39–46% on all four domains but costs −0.235 [-0.331, -0.145] on COCO, −0.134 [-0.167, -0.103] on Urban and −0.166 [-0.251, -0.095] on SUIM (Rural: −0.024 [-0.051, 0.002]). Partial sharing is no better: the rule selects $k = 0$; sharing even the first layer costs −0.089, and two or more layers collapse accuracy.

The mechanism is again presence. Under full sharing presence *saturates* (median 0.87

Urban, 0.90 SUIM, 0.98 COCO) and the share of present classes missed drops to exactly zero: the presence token can no longer say "absent", so every absent concept produces false positives. **Concept presence is decided inside the text-conditioned fusion encoder, from its first layer onward**; there is no concept-independent prefix to amortise. A further idea, precomputing the pixel decoder's lateral convolutions through linearity, saves nothing, since the concept-dependent term must still be convolved at full resolution for every concept.

7.7 Summary

SAM 3's cost is its image encoder (time) and text encoder (weights), not the decoder or fusion encoder the preliminary proposals targeted. Dynamic INT8 with a cached vocabulary and a one-time export gives SAM 3 in a 657 MB file and 2.5 GB of memory with no measurable accuracy loss on 400 unseen images, and with no data or training. The existing small model, EfficientSAM3, is fast but loses more than half of SAM 3's accuracy, and the loss is traced to its image encoder collapsing the presence estimate. Sharing per-concept computation fails through the same component. Chapter 8 uses this diagnosis to train a better small model.

Chapter 8

Output Distillation Through the Frozen Concept Path

Chapter 7 located EfficientSAM3's accuracy loss in its small image encoder and the mechanism in presence: a backbone trained to imitate SAM 3's features produces features the presence head rejects. EdgeSAM [51] found for SAM 1 that imitating encoder features is insufficient and that distilling through the prompt-conditioned decoder recovers accuracy. This chapter answers RQ5 by applying that principle to SAM 3's text-prompted *concept* path, where the teacher's presence is an explicit training target, and by confirming the result on 400 images never used before.

8.1 Method

8.1.1 Student and teacher

The student is EfficientSAM3's released Stage-3 image backbone (EfficientViT-B1 with its neck, 22.1 M parameters). It feeds SAM 3's own frozen fusion encoder, decoder and heads, with SAM 3's text features for the concept. The teacher is the same frozen stack fed by SAM 3's ViT (Figure 8.1). Before training, this starting point scores 0.303 on COCO test, matching the released EV-M model (0.332), which uses its own fine-tuned text encoder, and it shares EV-M's false-positive problem (1.83 per true positive).

8.1.2 Objective

For an image and a concept, let p^T, p^S be teacher and student presence probabilities, q_i^T, q_i^S the 200 query probabilities, m^T, m^S the semantic maps, and b_i, M_i box and mask outputs. With teacher probabilities as soft targets,

$$\begin{aligned}\mathcal{L} = {} & 2\,\mathrm{BCE}(p^S, p^T) + \frac{1}{200}\sum_i \mathrm{BCE}(q_i^S, q_i^T) + \mathrm{BCE}(m^S, m^T) \\ & + \sum_{i\in\mathcal{K}} w_i\big(\|b_i^S - b_i^T\|_1 + \mathrm{BCE}(M_i^S, M_i^T) + \mathrm{Dice}(M_i^S, M_i^T)\big),\end{aligned} \tag{8.1}$$

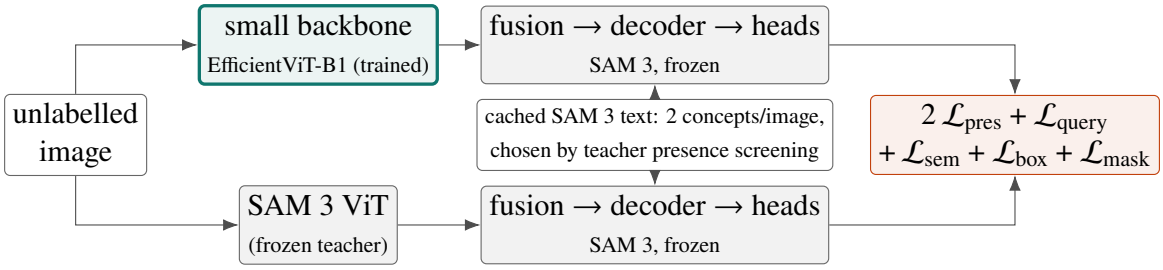


Figure 8.1: Concept-in-the-loop output distillation. One frozen copy of SAM 3 is shared; only the image backbone differs between teacher and student. Both sides run the same frozen decoder with the same 200 learned queries, so outputs are compared query by query without matching. No labels are used.

where $\mathcal{K}$ is the teacher's top-10 queries ranked by final score $s_i = \sigma(\ell_{q,i})\sigma(\ell_p)$ and w_i is that score, normalised. Presence has weight 2 because it is the measured failure.

8.1.3 Data and training

Training images are 4,000 COCO train2017 images whose annotations are never read; they are disjoint from every val2017 evaluation image. Batch normalisation statistics are frozen. The optimiser is AdamW (weight decay 0.01) with 50-step warm-up, cosine decay and gradient clipping at 1.0; one step is 8 images with 2 concepts each, in bfloat16. Fitting two copies of the SAM 3 heads plus a 446 M teacher ViT on an 8 GB GPU required activation checkpointing through the frozen stack (peak 5.9 GB, about 22 s per step). Checkpoints are written atomically every 25 steps so that training resumes after interruption; the workstation lost mains power several times during the experiments without loss of finished work.

8.2 Pilot 1: Uniform Concept Sampling Fails

Sampling two of the 80 COCO class names uniformly makes about 95% of sampled concepts absent from the image (mean teacher presence 0.05). With learning rate 10^{-4} the training loss fell (2.14 → 1.84), but test mIoU dropped from 0.303 to 0.159 after 300 steps (−0.144 [-0.262, -0.012]). The failure was the opposite of the starting point's: false positives fell from 1.83 to 0.57 per true positive while the share of present classes missed rose from 20% to 58%. **The student learned "absent"**—a lower loss that vetoes real objects. The worst checkpoint coincided with the learning-rate peak, so the rate was also too aggressive for an already-trained backbone.

8.3 Pilot 2: Teacher-Balanced Concept Screening

For each image the frozen teacher scores 24 random class names without gradients, and the student trains on the most-present one plus one other. This balances present and absent

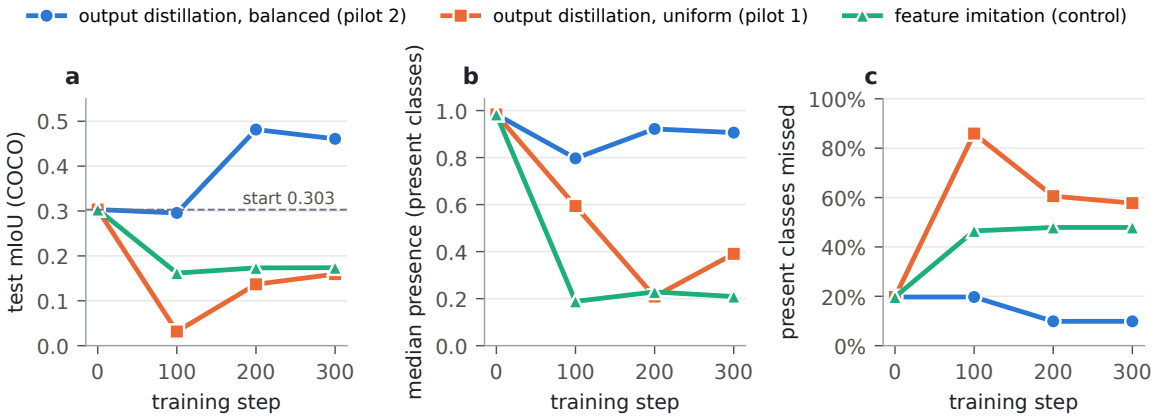


Figure 8.2: The first 300 steps on COCO (test mIoU at dev-selected thresholds; presence and missed classes on test). Output distillation with teacher-balanced concepts (pilot 2) improves; uniform sampling (pilot 1) learns to veto; feature imitation on identical data degrades with the collapsing-presence signature of EfficientSAM3's Stage 1.

concepts without labels: over the run the kept class was present (teacher presence > 0.5) in 61% of images, and mean teacher presence on trained concepts rose from 0.05 to 0.32. The learning rate was lowered to 2×10^{-5}. A go/no-go gate fixed in advance required dev mIoU at step 100 to exceed the start (0.301); it reached 0.353.

8.4 Control: Feature Imitation on the Same Data

Any gain could come from the objective or merely from 2,400 extra images. An identical run therefore trains the backbone only to match SAM 3's three FPN feature maps (scale-normalised mean squared error), with the same initialisation, images, steps and learning rate.

Figure 8.2 shows the three runs. After 300 steps (2,400 unlabelled images, about 110 minutes), pilot 2 reached 0.461 on the 50 test images, with fewer false positives (1.83 → 1.12 per true positive) and half as many present classes missed (20% → 10%). The control *degraded* to 0.16–0.17, with median presence about 0.2 and nearly half of present classes missed. Output distillation beat the control by +0.299 [+0.149, +0.415].

8.5 Long Run

The pilot-2 recipe was run unchanged for 1,000 steps (8,000 images, two epochs; about 5.5 h of training), and every 100-step checkpoint was evaluated on the 15-image dev tier and the 50-image test tier.

Figure 8.3 exposed a problem with the evaluation rather than with training. Under the fixed threshold 0.3 the final checkpoint scored 0.606, with a flat plateau over the last three checkpoints. Under dev-selected checkpoint and threshold, dev picked step 200 (0.421 on

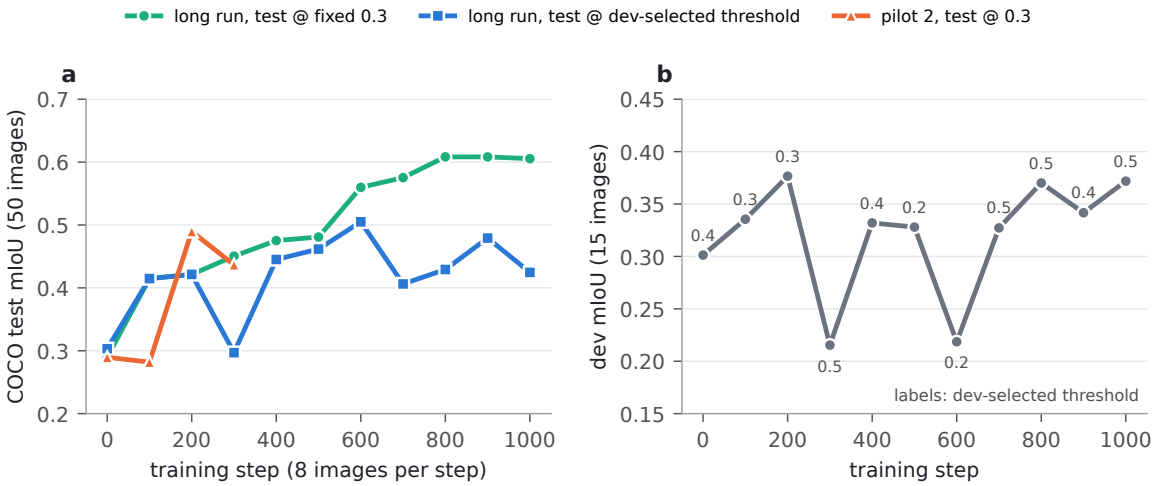


Figure 8.3: The 1,000-step run on the original small tiers. (a) Test mIoU at the fixed threshold 0.3 rises to a plateau near 0.61; under dev-selected thresholds it does not. (b) Dev mIoU swings by up to 0.16 between adjacent checkpoints, and the dev tier selects thresholds (labels) that do not match where test curves peak: 15 images are too few to select with.

test), which would suggest that longer training did not help. The two protocols disagreed because the 15-image dev tier was too small: it selected thresholds of 0.4–0.5 for late checkpoints whose test curves all peaked at 0.3. Choosing the fixed-threshold reading *after* seeing this would itself be a selection, so neither number was accepted; instead, a larger evaluation set was built with its analysis plan fixed in advance.

8.6 Confirmation on 400 Unseen Images

Plan (written before any model was evaluated on the set). 500 COCO val2017 images disjoint from all 90 used so far, each with at least one non-crowd instance ($\geq 32 \times 32$ pixels) of the five classes, drawn round-robin across classes and split by seed into dev 100 and test 400 (Table 4.2). The *primary* result is the long run's final checkpoint (step 1000, not chosen by any score) at the threshold selected on the 100-image dev tier, compared by paired cluster bootstrap (4,000 resamples) against released EfficientSAM3 EV-M and TV-M, the starting point, SAM 3 and INT8 SAM 3, each at its own dev-selected threshold. *Secondary* analyses: the same comparisons at the fixed threshold 0.3; steps 800 and 900 (stability); pilot 2 versus the control.

Results. Figure 8.4 and Tables 8.1 and 8.2 give five conclusions.

1. **Output distillation beats the released small models.** The final checkpoint scores 0.486 against 0.339 for EV-M (+0.147 [+0.113, +0.184]) and 0.386 for TV-M, at identical architecture, size and speed to EV-M. It also produces about a third as many false positives per true positive.

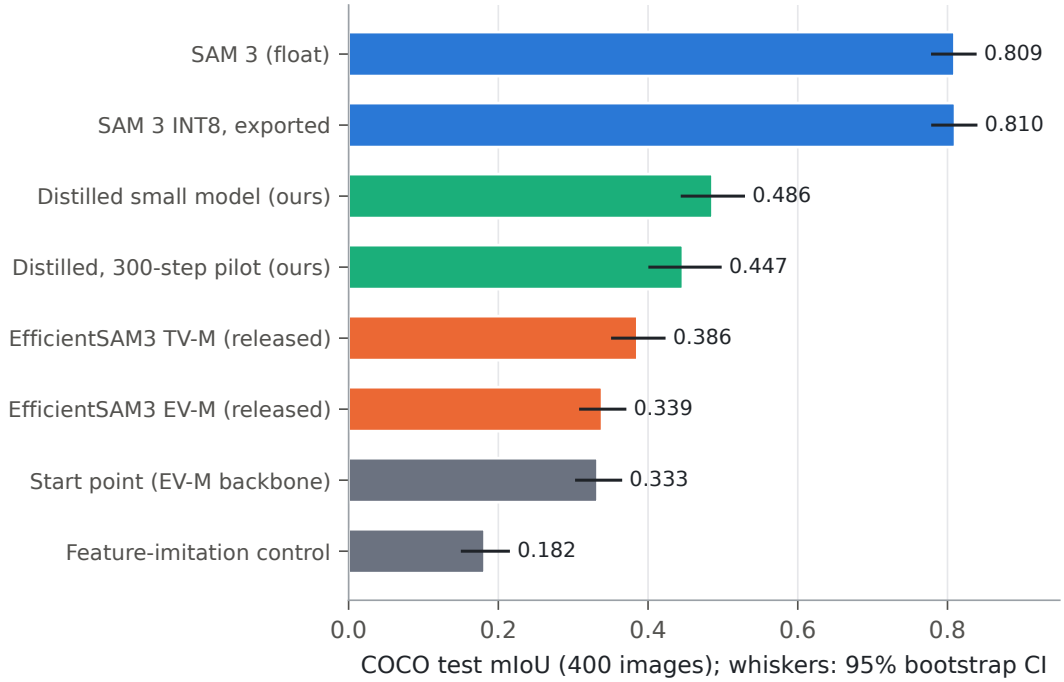


Figure 8.4: Confirmation set: COCO test mIoU on 400 unseen images, each model at the threshold selected on 100 separate dev images, with 95% bootstrap intervals.

2. **The gain is caused by the objective, not the data.** On the same images, pilot 2 beats the feature-imitation control by +0.265 [+0.215, +0.317].
3. **Longer training helps a little and is stable.** The final checkpoint is +0.039 above pilot 2 at the fixed threshold, and steps 800–1000 agree (0.475–0.486). With 100 dev images the selected threshold (0.3) coincides with the fixed one, so the protocol ambiguity of Section 8.5 disappears.
4. **The 50-image estimate was optimistic.** The long run's 0.606 on the original test tier is 0.486 on 400 new images. Adjacent checkpoints evaluated on the same 50 images are strongly correlated, so their agreement was not independent evidence. About one third of the gap between EfficientSAM3 and SAM 3 is closed, not more than half.
5. **INT8 SAM 3 loses nothing measurable.** The exported INT8 model scores 0.810 against 0.809 for float SAM 3 (+0.001 [-0.004, +0.006]; at threshold 0.3, +0.000 [-0.004, +0.005]). The −0.025 seen on the 50-image tier (Table 7.6) was a small-sample effect concentrated in one class.

Per class (distilled / EV-M / SAM 3): person 0.67 / 0.38 / 0.84, car 0.45 / 0.30 / 0.74, bird 0.25 / 0.11 / 0.85, dog 0.62 / 0.59 / 0.86, boat 0.43 / 0.31 / 0.75. The weakest class is now bird: small, thin objects.

8.7 Deployment Summary

Figure 8.5 places the options on one plane.

Table 8.1: Confirmation results on 400 test images. Thresholds selected on 100 dev images. FP/TP: false positive instances per true positive; missed: share of present classes with no surviving instance.

Model	thr	test mIoU	95% CI	@0.3	FP/TP	missed
SAM 3, float32	0.6	0.809	[0.780, 0.837]	0.779	0.08	5%
SAM 3, INT8 exported	0.6	0.810	[0.780, 0.838]	0.780	0.08	5%
Distilled, final (step 1000)	0.3	**0.486**	[0.446, 0.527]	0.486	0.44	18%
Distilled, step 900	0.3	0.486	[0.445, 0.528]	0.486	0.41	19%
Distilled, step 800	0.4	0.475	[0.433, 0.521]	0.479	0.37	22%
Distilled, pilot 2 (step 300)	0.3	0.447	[0.402, 0.496]	0.447	0.65	17%
EfficientSAM3 TV-M (released)	0.6	0.386	[0.353, 0.421]	0.287	1.07	24%
EfficientSAM3 EV-M (released)	0.4	0.339	[0.310, 0.369]	0.313	1.44	16%
Starting point (step 0)	0.4	0.333	[0.305, 0.363]	0.308	1.61	15%
Control, feature imitation	0.1	0.182	[0.152, 0.213]	0.112	0.35	48%

Table 8.2: Pre-registered comparisons on the 400 confirmation images (paired cluster bootstrap).

Role	Comparison	Δ mIoU [95% CI]
Primary	distilled (final) − EfficientSAM3 EV-M	+0.147 [+0.113, +0.184]
Primary	distilled (final) − EfficientSAM3 TV-M	+0.100 [+0.060, +0.141]
Primary	distilled (final) − starting point	+0.154 [+0.120, +0.190]
Primary	distilled (final) − SAM 3	-0.323 [-0.362, -0.282]
Primary	distilled (final) − SAM 3 INT8	-0.324 [-0.364, -0.282]
Secondary	at fixed 0.3: distilled − EV-M	+0.173 [+0.135, +0.213]
Secondary	at fixed 0.3: distilled − pilot 2	+0.039 [+0.003, +0.074]
Control	pilot 2 − feature imitation	+0.265 [+0.215, +0.317]
Deployment	SAM 3 INT8 − SAM 3 float	+0.001 [-0.004, +0.006]

- **When accuracy matters and 2.5 GB of memory is available**: INT8 SAM 3 with a cached vocabulary matches float SAM 3 (0.810 versus 0.809) in a 657 MB file, with no data or training.
- **When speed matters**: the distilled student runs at EV-M's cost (about 6.5 s for a five-concept image on CPU, 2.6× faster than float SAM 3 and 1.7× faster than INT8 SAM 3) and is substantially more accurate than any released small SAM 3.

Neither route reduces the per-concept cost of about 1 s on CPU, which Section 7.6 showed cannot be amortised without destroying presence.

8.8 Summary

Distilling a small backbone through SAM 3's frozen concept path, with presence as an explicit target and concepts balanced by the teacher, recovers accuracy that feature imitation destroys. Confirmed on 400 unseen images under a plan fixed in advance, it lifts EfficientSAM3's backbone from 0.339 to 0.486 mIoU at unchanged cost, and a matched control attributes the gain to the objective: EdgeSAM's lesson holds for SAM 3's text-prompted

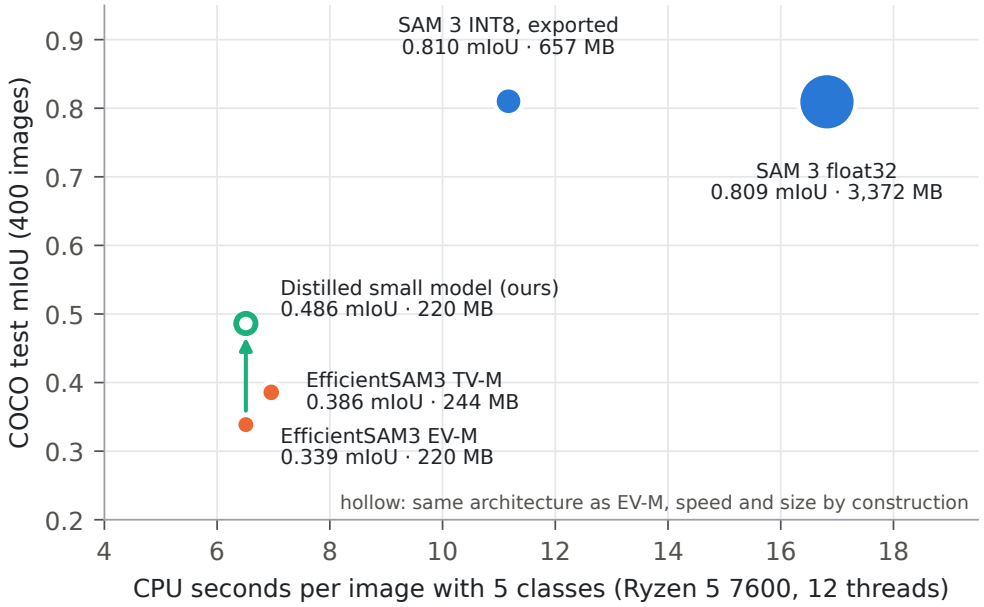


Figure 8.5: Deployable options on CPU: COCO test mIoU on the 400 confirmation images versus time for one image with five concepts; marker area is proportional to file size. The distilled student shares EV-M's architecture, so its latency and size are EV-M's by construction (hollow marker).

path, and the reason is presence.

Chapter 9

Discussion

This chapter draws the two parts of the thesis together, sets out what the project taught us about evaluating inference-time methods, and discusses ethical considerations and limitations.

9.1 Presence Is the Fragile Link in SAM 3

The domain-adaptation study and the efficiency study were designed independently. They converge on the same component (Table 9.1).

Two architectural facts make presence fragile. It is *multiplicative* (Equation (2.2)), so any bias in it moves every instance of a concept across the threshold at once. And it is decided *inside the text-conditioned fusion encoder*, from the first layer (Section 7.6), so it depends jointly on the image features and on the balance of the prompt's text features. A feature-imitation student whose features are close in mean squared error can still push the presence head into a regime it was never trained on; the gist/detail operator shows the same sensitivity from the text side, moving presence more than ten times as much as the best query score.

This suggests a practical rule for anyone adapting or compressing SAM 3: **measure presence first**. Its median on present classes and the share of present classes with no surviving instance diagnosed every failure in this thesis before any accuracy number was computed, at negligible cost.

9.2 Revisiting the Preliminary Hypotheses

Table 9.2 summarises how each preliminary hypothesis fared.

9.3 Lessons for Evaluating Inference-Time Methods

Check the metric on a case whose answer is known. A top-1 read-out of a multi-instance model scored against a class region is capped by construction. Correcting it changed mea-

Table 9.1: Every intervention in this thesis succeeded or failed through the presence score.

Intervention	What happened to presence	Consequence
Domain shift (baseline SAM 3)	Pessimistic: median 0.78–0.81 on present classes; 22–27% of classes vetoed	Correctly localised objects discarded
Gist/detail rebalancing (Chapter 6)	Lifted by 0.12–0.13 out of distribution; 0.02 on COCO	Rescued detections; gain only where the veto is active
Sharing the fusion encoder (Section 7.6)	Saturated (0.87–0.98); no class ever judged absent	False positives on absent concepts; up to −0.24 mIoU
Feature-imitation distillation (Section 7.4, control)	Collapsed (0.00–0.04 for Stage 1; about 0.2 for the control)	Near-zero accuracy; control falls from 0.30 to 0.16–0.18
Output distillation through presence (Chapter 8)	Restored (median 0.91 at step 300; missed classes halved)	0.339 → 0.486 on 400 unseen images

Table 9.2: Preliminary hypotheses and what the re-evaluation found.

Preliminary hypothesis	Verdict	Evidence
ConceptBank improves SAM 3 on aerial imagery (+0.019)	Not supported	+0.0001 under the corrected protocol; well-powered null (Chapter 5)
Visual prototypes inject domain knowledge	Rejected	Correct, shuffled, wrong-domain and absent prototypes perform identically
Blend fails because it leaves the text manifold	Rejected	Step-size artefact: $\rho = 0.51$ versus 0.034
Steering the text embedding can help	Supported, reframed	Data-free gist/detail rebalancing: +0.090 underwater, ≈ +0.02 aerial, 0 on COCO (Chapter 6)
DEQ / deformable attention make SAM 3 efficient	Not pursued	Targets 0.1–4% of compute; require retraining (Section 7.2)
SAM 3 can be made practical on low-end hardware	Supported	INT8 in 2.5 GB; distilled student 0.486 at EV-M cost (Chapters 7 and 8)

sured mIoU by +0.258, fourteen times the effect under study. Published zero-shot SAM numbers on remote sensing that use single-mask read-outs are likely understated for the same reason.

Compare operators at matched perturbation size. The claim that blending destroys the text manifold while residual steering preserves it compared a relative step of 0.51 with one of 0.034. Parameterising every operator by ρ and testing against random directions of the same norm turned that claim, and the claim that prototypes carry domain knowledge, into testable statements; both failed.

Thresholds are hyperparameters. In a thresholded pipeline a shared fixed threshold can manufacture or hide a gain, and selecting each arm's best threshold on test data inflates it in the other direction. Re-selecting inside every bootstrap replicate reports both honestly at

little cost.

Evaluation sets must be large enough to select with. A 15-image dev tier could not choose a checkpoint or threshold, and a 50-image test tier overstated the distilled student (0.606 versus 0.486 on 400 images). Adjacent checkpoints evaluated on the same images are not independent confirmations.

Pre-registration protects against oneself. Four results in this project looked convincing and were wrong or overstated: a presence gate tuned on test data, a "ratio law" whose strongest evidence holds by construction, an aerial gain of +0.038 whose reliable size is about +0.02, and the 50-image distillation estimate. They were caught by re-running with every choice fixed in advance, by reading the operator's code, by pricing selection into intervals, and by a confirmation set with a written plan. None would have been caught by a single headline number.

9.4 Ethical Considerations

Dual use. Segmentation of aerial imagery and efficient deployment on drones and embedded boards serve environmental monitoring, disaster response and urban planning, but the same capabilities can support surveillance. This thesis releases no new capability beyond making an openly released model smaller and measuring it; we nonetheless note that lowering the hardware bar widens access for all uses.

Reliability and over-claiming. A central finding is that a published-looking improvement, including our own, can be an evaluation artefact. Deploying a method on the strength of such a number in a safety-relevant setting would be harmful. We therefore report nulls, corrections and withdrawn hypotheses alongside positive results, and state the evaluation conditions of every number.

Data. All datasets are public research benchmarks (LoveDA, SUIM, COCO) used under their licences. Distillation used unlabelled COCO training images; no personal data were collected. COCO images contain people, and a model distilled on them inherits whatever biases SAM 3 and COCO carry; we did not audit these.

Environmental cost. All training ran on one 8 GB consumer GPU (roughly 10 h for all distillation runs), and the thesis's efficiency results aim to reduce the energy needed to use the model.

9.5 Limitations

- **Sample sizes in Part I.** Domain-adaptation test tiers have 50 images per domain and 5–6 classes. They resolve the large effects (protocol, controls, SUIM) but not the aerial gain of about 0.02 with confidence. A dev tier for the recalibration operators was run only on LoveDA Urban; the underwater threshold rule was transferred rather than selected on SUIM.
- **One architecture.** All findings concern SAM 3. The presence mechanism is specific to presence-factored detectors, and the gist/detail result may not transfer to models that consume text features differently.
- **Reconstruction.** The preliminary implementation was not preserved; our rebuild follows the preliminary report and design notes and reproduces its baseline (0.2450 versus 0.2373), but cannot explain the originally reported forest score of 0.000.
- **Efficiency measurements.** Latency and memory were measured on a 12-thread desktop CPU, not an edge device. The exported INT8 model and the distilled student were not separately latency-benchmarked; their costs follow from bit-identical computation and identical architecture respectively. Dynamic INT8 was the only quantisation scheme tried, and a PyTorch runtime overhead of about 0.7 GB remains.
- **Distillation scope.** The student was trained and evaluated on COCO only, with one backbone (EfficientViT-B1). It remains about 0.32 mIoU below SAM 3.

Chapter 10

Conclusion and Future Work

10.1 Conclusion

This thesis set out to make SAM 3 useful beyond ordinary photographs and ordinary hardware, starting from two proposals of our own preliminary phase: ConceptBank for domain adaptation and deep-equilibrium and deformable architectures for efficiency. Rebuilt and re-tested under a pre-registered protocol with matched controls, neither proposal held as stated, and what replaced them is better supported and more useful.

Domain adaptation (RQ1, RQ2). ConceptBank's reported gain does not reproduce, its prototypes contribute nothing, and its evaluation protocol had halved SAM 3's measured accuracy. What remains is a data-free rebalancing of a prompt's gist against its detail. It improves underwater segmentation by +0.090 mIoU out of sample, aerial segmentation by a reliable +0.02, and natural photographs not at all. The gain consists of objects SAM 3 had localised but vetoed through a pessimistic presence estimate, and the prediction that follows—no gain where presence is already confident—holds on COCO.

Low-end deployment (RQ3–RQ5). Measured rather than counted, SAM 3's cost lies in its image and text encoders, not in the components the preliminary proposals targeted. Dynamic INT8 with a cached vocabulary, exported once, runs SAM 3 in 2.5 GB of memory and a 657 MB file with no measurable loss of accuracy on 400 unseen images. The existing small SAM 3, EfficientSAM3, loses more than half of SAM 3's accuracy because its distilled image encoder collapses presence, and sharing per-concept computation fails through the same component. Distilling a small backbone through SAM 3's frozen concept path, with presence as an explicit target and teacher-balanced concepts, lifts it from 0.339 to 0.486 mIoU on 400 unseen images at unchanged cost, and a matched control attributes the gain to the objective.

Across both parts, one conclusion stands out: **for SAM 3, whether a change helps is largely a question of what it does to presence.**

10.2 Future Work

1. **Confirm the recalibration at scale.** Run the amplify-detail operator on LoveDA Urban's more than 450 never-evaluated held-out images and on dev tiers for SUIM and COCO, with thresholds selected on dev.
2. **Presence-aware calibration.** Separate "present but vetoed" from "truly absent" using information beyond the presence head, for instance agreement between query scores and presence, which gating on presence alone cannot do.
3. **Stronger students.** Longer output distillation, a larger backbone (TinyViT-11M), unlabelled target-domain images for aerial and underwater deployment, and INT8 export of the distilled student.
4. **The per-concept cost.** About 1 s per concept on CPU remains and cannot be amortised by sharing the fusion encoder. Reducing it without touching concept conditioning—lighter pixel decoding, fewer fusion layers learned by output distillation, or skipping mask heads for concepts with confidently low presence—is the open efficiency problem.
5. **Edge hardware and runtimes.** Benchmark on an ARM board or a laptop CPU, and move to a non-PyTorch runtime to remove about 0.7 GB of framework overhead.

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Appendix A

Proofs

Proof of Proposition 5.1. Write $a = -\bar{E}$ and $\delta = a + P$ with $\|P\| = \kappa\|a\|$, $\kappa < 1$. For fixed $\|P\|$ the angle ϕ between a and $a + P$ is maximised when $a + P \perp P$, in which case a, P and $a + P$ form a right triangle with hypotenuse a, so $\sin\phi = \|P\|/\|a\| = \kappa$. Hence $\phi \leq \arcsin\kappa$ for every P of that norm. □

Proof of Proposition 6.1. Over valid tokens $\sum_t R_t = \sum_t (E_t - \bar{E}) = 0$. Hence $\langle \bar{E}\mathbf{1}^\top, R\rangle_F = \langle \bar{E}, \sum_t R_t\rangle = 0$, and $\|\alpha\bar{E}\mathbf{1}^\top + \beta R\|_F^2 = \alpha^2|\mathcal{V}|\|\bar{E}\|^2 + \beta^2\|R\|_F^2$. Mean-shrink subtracts $s\bar{E}$ from every valid token, i.e. $\alpha = 1 - s$, $\beta = 1$, with $\|\Delta\|_F = s\sqrt{|\mathcal{V}|}\|\bar{E}\| = s\,g\,\|E\|_F$; the realised step is $\rho = sg$. For the renormalised operator, $E' = \|E\|_F\,(\alpha\bar{E}\mathbf{1}^\top + \beta R)/\|\alpha\bar{E}\mathbf{1}^\top + \beta R\|_F$, which is invariant under $(\alpha, \beta) \mapsto (c\alpha, c\beta)$ for $c > 0$ and therefore depends on $r = \alpha/\beta$ only. □

Appendix B

Engineering Details

B.1 Pitfalls That Silently Change Results

The following defects would not raise errors but would have changed the numbers. Each applies to any work that evaluates or modifies SAM 3 at inference time.

Pitfall	Consequence and fix
`forward_grounding` mutates the dictionary of cached backbone outputs it receives	Results would depend on operator order. A shallow copy is passed; a test asserts cached features are bit-identical across ≈400 calls.
SAM 3's text mask marks *padding* as `True`	Pooling over the wrong tokens corrupts $\bar{E}$, ρ and every operator. All pooling goes through one inversion.
`masks_logits` are sigmoid probabilities, not logits	Thresholding twice would shrink masks; conversely, threshold sweeps are free.
The upstream batched image+text path mis-attributes captions	Custom batched path; a test asserts batched and sequential outputs agree.
LoveDA label PNGs are palette images	Converting to RGB destroys class ids; masks are read as indices.
Spectral clustering kept eigenvectors "at the smallest non-zero eigenvalues"	With c components there are c zero eigenvalues; the Ng–Jordan–Weiss form [31] is used (pinned by a test).
Operator-precedence typo in the control filter (`a or 0 - b`)	Admitted 41 control seeds instead of 19; found by printing the control count.
Building INT8 SAM 3 in memory peaks at 7.8 GB	Two model processes hung a 14 GB machine; a guard refuses a second build, and deployment loads an exported model.
EfficientSAM3's builder loads LiteText context-16 checkpoints before truncating the text context	Checkpoints fail to load; the context is truncated first.
A pickled text cache referred to a class defined in `__main__`	The exported model failed to load elsewhere; the class lives in an importable module.

Pitfall	Consequence and fix
Exported model evaluated with a quantized text encoder, cache built from float	Deployed and in-memory models disagreed; one shared preparation function now builds both.
Uniform concept sampling during distillation	95% absent concepts taught the student to veto (pilot 1); teacher-balanced screening fixed it.
Host memory growth during long distillation	The process was killed by the operating system; allocator trimming, fewer data workers and resumable checkpoints fixed it.

B.2 Exactness Tests for the Efficiency Code

Table B.2: Exactness tests (`tests/test_lowend_grounding.py` and `tests/test_lowend_deploy.py`).

Test	Max. abs. difference
Shared-memory path fed each concept's own memory vs. `forward_grounding`	0.0
Three concepts batched vs. one at a time (shared path)	1.8×10^{-6}
Partial sharing, $k = 0$, vs. `forward_grounding`	0.0
Partial sharing, any k, "empty" prompt equal to the concept prompt	0.0
Partial sharing, $k = 6$, vs. full sharing	0.0
Exported INT8 model vs. in-memory INT8 model (logits, boxes, masks, presence, text)	0.0
Cached text features: batch of five names vs. one at a time	1.8×10^{-5}

B.3 Distillation Hyperparameters

Table B.3: Output-distillation settings (pilot 2 and the long run; the control differs only in its loss).

Setting	Value
Student initialisation	EfficientSAM3 Stage-3 EfficientViT-B1 + neck (22.1 M parameters)
Frozen components	SAM 3 ViT (teacher), fusion encoder, decoder, heads, text features
Training images	4,000 COCO train2017, annotations unused
Concepts per image	2, from teacher screening of 24 random class names
Images per step	8 (gradient accumulation)
Steps	300 (pilots, control); 1,000 (long run)
Optimiser	AdamW, lr 2×10^{-5} (pilot 1: 10^{-4}), weight decay 0.01
Schedule	50-step linear warm-up, cosine decay; gradient clipping 1.0
Loss weights	presence 2; query, semantic, box, mask 1; top-10 teacher queries
Precision and memory	bfloat16; activation checkpointing; BatchNorm frozen; peak 5.9 GB
Hardware and time	RTX 4060 8 GB; $\approx$22 s per step

Appendix C

Reproducibility

Every table and figure is regenerated by script from per-image records that store raw TP/FP/FN for every (image, class, operator, threshold).

```
# domain-adaptation study (about 1 h on an RTX 4060)
bash experiments/run_gpu.sh
python -m conceptbank.cli analyze \
    --run experiments/runs/main_urban_gpu --protocol union

# low-end: accuracy of any model, CPU benchmark, INT8 export
python -m lowend.run --model evm --config configs/lowend/coco_identity.yaml \
    --splits configs/splits/coco.json --device cuda --autocast bfloat16
python -m lowend.bench --model teacher_fork --quant int8_dynamic --threads 12
python -m lowend.deploy export --model teacher_fork --quant int8_dynamic \
    --vocab person car bird dog boat --out experiments/deploy/sam3_int8_coco5

# output distillation (pilot 2), its control, and the long run
python -m lowend.distill --init stage3 --steps 300 --accum 8 --concepts 2 \
    --screen 24 --lr 2e-5 --out experiments/distill/pilot2_balanced
python -m lowend.distill --init stage3 --steps 300 --accum 8 --concepts 2 \
    --lr 2e-5 --feature-only --out experiments/distill/control_feature_only
bash experiments/run_distill_long.sh

# confirmation set (500 images) and its evaluation
python -m lowend.build_coco_large --n-dev 100 --n-test 400 --seed 1
bash experiments/run_cocolarge_eval.sh

# figures and numbers for this thesis
python docs/paper/make_figures.py
python docs/thesis/make_thesis_figures.py
pytest -m "not slow"
```

Bootstrap intervals use 2,000–10,000 paired resamples (seed 12345; 4,000 for the distillation comparisons). Re-running with a different count moves interval endpoints by at most 0.004 and changes no conclusion.